

Metro Station functional clustering and dual-view recurrent graph convolutional network for metro passenger flow prediction

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ABSTRACT

The metro system is indispensable for alleviating traffic congestion in the urban transportation system. Precise metro passenger flow (MPF) prediction is crucial in ensuring smooth operations of the metro system. Recently, the graph convolutional network (GCN), which is effective in the spatial feature extraction, has been applied in traffic prediction. However, most existing GCN-based methods construct the empirical graphs based on distance and adjacency, which cannot fully express the correlations of metro stations. This paper proposes a novel MPF prediction method consisting of three parts: K-means-based metro station functional clustering (KMSFC), external feature fusion, and dual-view recurrent GCN (DVRGCN). The KMSFC identifies the metro stations both having similar MPF changing tendencies and being located in similar urban functional areas. Furthermore, the DVRGCN is designed to simultaneously capture the spatiotemporal and external features. The dual-view GCN module in the DVRGCN captures both explicit and implicit spatial features of the metro traffic network. To demonstrate the capability for making accurate MPF predictions, the experiments using a real-world metro traffic dataset are conducted. The ablation experiments are also performed to prove the contribution of each module in the proposed method. The experimental results show that the proposed method outperforms other state-of-the-art traffic prediction methods.

1. Introduction

The metro system plays an important role in urban transportation system, increasing urban traffic carrying capacity and alleviating road traffic congestion. An increasing number of the citizens commute by metro because of the transportation rapidity, reliability, accessibility, and sustainability (Zhang, He, Bao, Hong, & Shi, 2020). However, the rising demand brings the rapidly growing throughput of passengers for the metro system, further causing severe congestion in those heavy-traffic metro stations (e.g. the junction station), especially during the rush hour (He, Wang, Jiang, Ma, & Wang, 2021). A precise prediction of the metro passenger flow (MPF) is crucial for congestion reduction and relevant planning of station redevelopment (Wang et al., 2021). Besides, passengers can schedule their trips in advance for avoiding congestion (Zhang, Chen, Guo, & Li, 2020) according to the precise MPF prediction information. For those shops around metro stations, precise MPF predictions can also facilitate their business-hour planning. During the low-MPF periods, the owners could choose to temporally

shut the shops to reduce the operating costs. However, complex traffic networks and variable travel behavior make traffic prediction an extremely difficult task. The MPF prediction can be regraded as a subclass of traffic prediction, and the effective modeling of intricate spatiotemporal patterns of traffic data is necessary for traffic prediction (Tedjopurnomo, Bao, Zheng, Choudhury, & Qin, 2022). Traditional traffic prediction methods include the statistical methods and machine learning (ML) methods. The statistical methods such as the autoregressive integrated moving average (ARIMA) (Box & Pierce, 1970) and its variants (Williams, 2001; Williams & Hoel, 2003) are effective to capture the variation of traffic time series. The ML methods, like the Bayesian network (BN) (Sun, Zhang, & Yu, 2006) and support vector regression (SVR) (Chen, Liang, Hong, & Gu, 2015; Wu, Ho, & Lee, 2004), are more flexible to learn the nonlinear patterns of traffic data. However, the applicability of these methods depends on either some ideal assumptions about data or manual features, so they may be

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insufficient to cope with the complex spatiotemporal dynamics of traffic data (Yin et al., 2022). In recent studies of traffic prediction, the deep learning (DL) methods have been shown the capability to outperform the aforementioned traditional approaches (Shi, Feng, Geng, Tang, & Wang, 2019). Among the DL models, the graph convolutional network (GCN) based model has become one of the most popular approaches to learn the spatiotemporal patterns of traffic data. The GCN-based modules have been effectively employed to aggregate information from different nodes of the traffic network, thanks to the capability of dealing with no-Euclidean traffic data (Guo, Lin, Feng, Song, & Wan, 2019; Li, Yu, Shahabi, & Liu, 2017; Zhao et al., 2020). The GCN requires the predefined graphs with prior knowledge of traffic networks to effectively extract the spatial features, and previous studies have designed the graphs based on adjacency, distances, population flow and other features (Chen, Fu, & Wang, 2022; Huang et al., 2022; Liu et al., 2022). However, in the MPF prediction, these characteristics could be insufficient to reflect the MPF changing trends. In this paper, the K-means-based metro station functional clustering (KMSFC) is proposed to automatically match the similarities of MPF time series and achieve metro station clustering. Generally, the metro stations located in the same urban functional area (e.g. residential area, business district, and school district) have similar passenger flow variation trends. Based on the KMSFC results, the passenger travel behaviors of metro stations within the same urban functional areas would be found. Furthermore, the results would be used to generate spatial prior knowledge graph.

Currently, the MPF data is mainly collected by the automatic fare collection (AFC) system (Tang, Zhao, Cabrera, Ma, & Tsui, 2018). The AFC data implicitly indicates the trace of inflow and outflow passengers, and the passenger inflow and outflow of the metro station also have different flow changing tendencies. Therefore, a well-defined graph of the metro system for the GCN should consider the stations that are not adjacent but have strong correlations. Besides, different patterns of inflow and outflow should also be taken into consideration. In order to make precise traffic prediction, capturing the temporal patterns of traffic data is also significant. The RNN, as a classical DL method of learning the temporal features, suffers from the vanishing gradient problem, which leads its bad performance on modeling the long-term dependence (Liu, Liu, & Jia, 2019). The long short-term memory (LSTM) and gated recurrent unit (GRU), as the evolution version of the RNN, have been widely used in traffic prediction (Hao, Lee, & Zhao, 2019; Ma, Zhang, Du, Ding, & Sun, 2019; Sun, Boukerche, & Tao, 2020). Recently, the temporal convolution layer (TCN) along with the GCN was adopted to extract the spatiotemporal features in a time span (Fang, Zhang, Meng, Xiang, & Pan, 2019; Wu, Pan, Long, Jiang, & Zhang, 2019). However, these DL models usually separate the spatiotemporal features extraction process (Lau & Wong, 2021). Besides the difficulty of modeling the complex spatiotemporal features, the MPF prediction is also confounded by the sudden increase of passenger flow during peak hours (Hao et al., 2019). Moreover, metro stations have different levels of MPF volume. The DL model should be robust to recognize the patterns of MPF variations in different metro stations, ever at peak times. To deal with the aforementioned research issues, this paper proposes an novel approach which consists of the KMSFC, external feature fusion, and dual-view recurrent GCN (DVRGCN) to accurately predict both the metro passenger inflow and outflow. The main contributions of this paper are summarized as follows:

- To obtain groups of the metro stations located in similar urban functional zones, this paper proposes a novel two-step KMSFC algorithm. Based on the clustering results, the functional urban areas of metro stations can be identified. Furthermore, a novel adjacency matrix containing the full semantic information of the metro traffic network is also constructed. The matrix provides an explainable spatial prior knowledge graph for the DL model.
- The external features related to time and volume levels of the MPF are considered in this paper. They are all represented by the one-hot encoding and embedded into the proposed model through the fully-connected layers. The fusion of external features helps the proposed DL model learn the MPF changing patterns relatively effectively.
- The DVRGCN is proposed to extract simultaneously the spatial and temporal features of the MPF data. The GCN module consists of a GCN in the spectral domain and a self-adaptive supplemental diffusion convolution network. Through this GCN module, the model can learn the given structural information from the static pre-constructed graphs of the metro network and adaptively capture the hidden spatial features. The Seq2Seq architecture of the DVRGCN ensures that the model effectively captures the temporal patterns of the MPF data.
- A practical metro dataset is used to conduct the experiments, which show that the proposed model is superior to the state-of-the-art baseline models. In addition, the ablation experiments verify the contributions of certain modules to the proposed method.

The rest of this paper is organized as follows. Section 2 reviews the research work on the metro-station clustering algorithms and traffic prediction methods. The specific details of the proposed MPF prediction method are demonstrated in Section 3. In Section 4, the results of the comparison experiments using the real-world metro traffic dataset are illustrated. In addition, the ablation experiments are also carried out in the same section. Finally, the conclusions and future work are discussed in Section 5.

2. Related work

The passenger travel patterns of the metro traffic system could be found by investigating the variations of the MPF time series. It could be intuitive that the passenger travel patterns of the metro stations in different urban functional areas are dissimilar. Therefore, the clustering of the metro stations is performed mainly based on the MPF time series clustering. Relevant research on the time series clustering and the clustering methods for public transit data are reviewed in Section 2.1. Mining deeper spatiotemporal relations of the traffic network is a key of the DL model to make better traffic predictions. The MPF prediction considered as traffic prediction is one of the critical and hot research topics for the intelligent transport system all the time. The traffic prediction methods are mainly divided into two categories: the traditional methods and DL-based methods. In addition, the traditional methods can be further divided into the statistical methods and ML methods (Cao, Dai, Liu, Yang, & Huang, 2022). The corresponding traffic and MPF prediction studies are reviewed in Section 2.2.

2.1. Metro station clustering

Clustering the metro stations based on the similar MPF trends is helpful for understanding the urban functional areas of metro stations (Li, 2021). Thus, the metro-station functional clustering can be converted into the time series clustering problem. The time series clustering mainly focuses on the time series similarity measurement and the selection of appropriate clustering algorithms (Aghabozorgi, Shirkhorshidi, & Wah, 2015). The Euclidean distance, Mikowski distance, Pearson correlation coefficients, and dynamic time wrapping (DTW) (Berndt & Clifford, 1994) are the common ways to measure the similarity between two time series (Liao, 2005). There are six categories of clustering algorithms, including the partitioning, hierarchical, grid-based, model-based, density-based, and multi-step clustering, according to different clustering requirements (Aghabozorgi et al., 2015). Selecting an appropriate clustering algorithm to cluster the time series is significant. The K-means and Euclidean distance measurement were combined to achieve the time series clustering (Vlachos, Lin,

Keogh, & Gunopulos, 2003), and the K-means is adopted along with DTW measurement to cluster the multimedia data (Niennattrakul & Ratanamahatana, 2006). The hierarchical clustering was used for time series clustering (Keogh & Pazzani, 1998; Van Wijk & Van Selow, 1999). The development of the DL has facilitated the emergence of the data-driven time series representation learning for clustering (Lafabregue, Weber, Gañarski, & Forestier, 2022). By virtue of the strong pattern recognition ability of the convolutional neural network (CNN), the time series was converted into pictures and the curve of the time series was represented through the pixel values (Anand & Nayak, 2021). Then, the feature maps for clustering were generated by the residual neural network (ResNet) (He, Zhang, Ren, & Sun, 2016). A GRU-based Seq2Seq model was proposed to extract the time series features and generated the feature vectors (Lai, Zhong, Pan, Ng, & Lai, 2021). Followed by measuring the similarities of features obtained from the DL models, they achieved precise time series clustering. The metro stations located in the similar urban functional areas generally have close passenger travel patterns. Clustering the MPF time series reveals the implicit patterns and finds spatiotemporal relations of metro stations (Li, 2021; Wang, Chen, & Zhang, 2018). Some studies also focused on identifying the importance (He, Agard, & Trépanier, 2020) and the similarity (Zhu, Yin, Qu, & Wu, 2021) of the public transit. However, those studies involved correlation analysis of metro stations without considering their functional locations. The KMSFC proposed by this paper clusters the historical MPF, and the results contain the urban functional information of metro stations. Moreover, the KMSFC provides the explainable priors of the functional areas where metro stations are located and the MPF variation pattern recognition for the DL model to make accurate traffic predictions.

2.2. Traffic prediction methods

The traditional and DL traffic prediction methods are discussed in this subsection. In Section 2.2.1, the traditional traffic prediction methods adopting the statistics and ML are reviewed. The limitations of the traditional traffic prediction methods are also mentioned. Recently, the DL-based methods have dominated the pattern analysis of large-scale dataset. In the traffic research fields, many studies used the DL methods to capture the spatiotemporal patterns of large-scale traffic dataset, achieving successful results. The corresponding traffic prediction research work is reviewed in Section 2.2.2.

2.2.1. Traditional methods

Statistical methods observe the historical traffic data time series, learn the variation patterns, and make predictions. The history average (HA) (Smith & Demetsky, 1997), which is an easy and fast statistical method, takes the average of the data of past time intervals as the prediction. The ARIMA (Box & Pierce, 1970) and its variants (Williams, 2001; Williams & Hoel, 2003) are the most typical statistical models in traffic forecasting. The Kalman filtering model (Jiao, Li, Sun, Hou, & Ibrahim, 2016) is also a category of effective statistical methods for traffic prediction. However, the performance of the statistical methods in traffic prediction is lackluster. The traffic environment is complicated and the traffic data is naturally nonlinear. Statistical methods depend on the linearity assumption of data, which is inefficient in forecasting the future traffic. To overcome the deficiencies of statistical methods, relevant research adopted the ML-based methods to handle the nonlinear traffic data. The ML methods including the BN (Pascale & Nicoli, 2011; Sun et al., 2006), SVR (Chen et al., 2015; Wu et al., 2004), and support vector machine (SVM) with wavelet transform (Sun, Leng, & Guan, 2015) were proposed to learn the high-dimensional traffic data and make precise traffic predictions. However, the ML models are not flexible to extract the spatiotemporal features of large-scale traffic data, and the shallow architecture requires the pre-constructed manual features (He et al., 2021). The aforementioned weaknesses restrict the performances of the traditional methods on the complex and dynamic traffic data.

2.2.2. Deep learning methods

With the tremendous success in computer vision and nature language processing (LeCun, Bengio, & Hinton, 2015), the DL methods are robust in learning the different patterns of large-scale dataset. The DL methods were introduced to solve traffic problems such as traffic classification, traffic generation, and traffic prediction (Cao et al., 2022). In the temporal correlation modeling, the recurrent neural network (RNN) is a typical DL model which can memorize short-term flow changing patterns. Researchers (Ishak, Kotha, & Alecsandru, 2003; Ma, Yu, Wang, & Wang, 2015) exploited the RNN to make traffic prediction. To reinforce the long-term dependence modeling ability of the RNN, its variants, i.e. the LSTM (Hochreiter & Schmidhuber, 1997) and GRU (Chung, Gulcehre, Cho, & Bengio, 2014) were proposed. The GRU-based DL model was utilized to make traffic flow prediction (Zhang & Kabuka, 2018). The LSTM was employed to model different periodical patterns of the MPF and make accurate MPF predictions (Liu et al., 2019). Moreover, the Seq2Seq encoder-decoder model is adopted along with the LSTM unit to achieve the precise forecasting of the short-term MPF (Hao et al., 2019). The memory and selective forgetting ability lead the RNN-based model to preferentially catch the data patterns of critical time points. However, these models are only temporal-oriented and the spatial features of traffic data are ignored. To solve the deficiency, they (Ma et al., 2019; Peng et al., 2020) converted the MPF data into matrices and used the CNN and LSTM to obtain both the spatial and temporal information. A channel-wise attention mechanism was also proposed to guide the model to assign the attention weights to different channels of the metro flow feature maps (Zhang, Che, Chen, Ma, & He, 2021). However, the traffic data is intrinsically in non-Euclidean space while the CNN is not suitable for non-Euclidean data (Diao et al., 2022). The GCN (Kipf & Welling, 2016) is a relatively feasible approach to capture the spatial features of the traffic networks. By adopting the graph convolution to achieve the node information aggregation (Bruna, Zaremba, Szlam, & LeCun, 2013; Niepert, Ahmed, & Kutzkov, 2016), the GCN is capable of saving the original topology information and extracting the local spatial features of the traffic networks (Bai et al., 2019; Lee & Rhee, 2022). In the metro network, the metro stations can be explicitly considered as the nodes and the metro lines represent the edges. Therefore, the multi-layer GCNs were adopted to capture the spatial dependencies at the citywide level and predict the future passenger flow volumes (Han et al., 2019).

To stimulate the potential of the GCN module, some studies (Chen et al., 2022; Lu, Ding, Ji, Sze, & He, 2021) integrated the adjacency, travel distance, and population flow of the metro stations into a single comprehensive graph for constructing the adjacency matrices. Meanwhile, previous studies (Li et al., 2023; Liu et al., 2022; Zhang et al., 2022) calculated the correlation coefficient relationship of traffic flow series to find all closely related traffic stations. All the manipulations are aimed at correlating traffic nodes even though they are not adjacent to each other. However, the above methods ignore the similar traffic flow variation patterns of different traffic stations. Our proposed KMSFC can mine the passenger commute patterns of the metro stations, even if those stations stay far away from each other. Based on the clustering results, the DL models can readily make pattern recognition of metro passenger flow data, which contributes to accurate predictions. Pre-constructed static graphs cannot include complete spatial relations of the whole traffic network. Some studies (Guo et al., 2022a; Ta et al., 2022; Wu et al., 2019) made a trainable adjacency matrix for the DL model to adaptively learn the dynamic adjacency matrix and mine the hidden spatial dependencies in the traffic network. Adaptive spatiotemporal graph networks learn implicit spatiotemporal features, but these features are insufficient and not explainable. Based on both the explainable priors from the KMSFC and adaptive spatiotemporal feature extraction module, the proposed model effectively recognizes the passenger flow patterns of metro stations in the same cluster and utilizes the learned spatiotemporal features to make predictions.

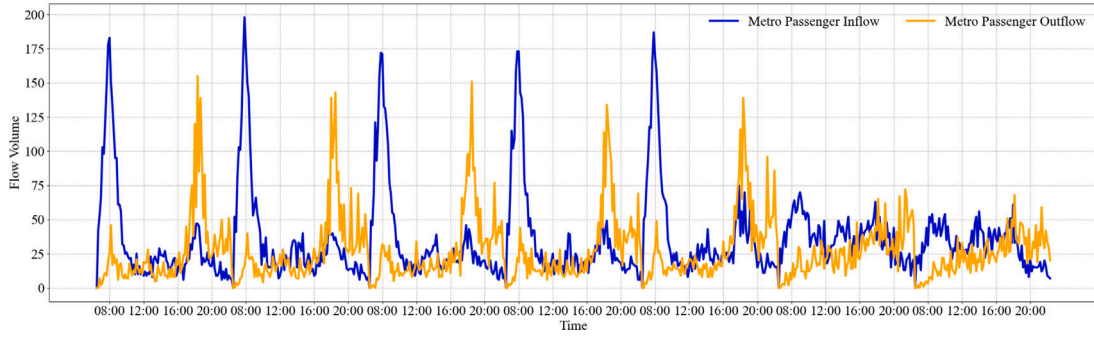


Fig. 1. Metro passenger flow in a week — Hongwan Station of Fuzhou Metro Line 2.

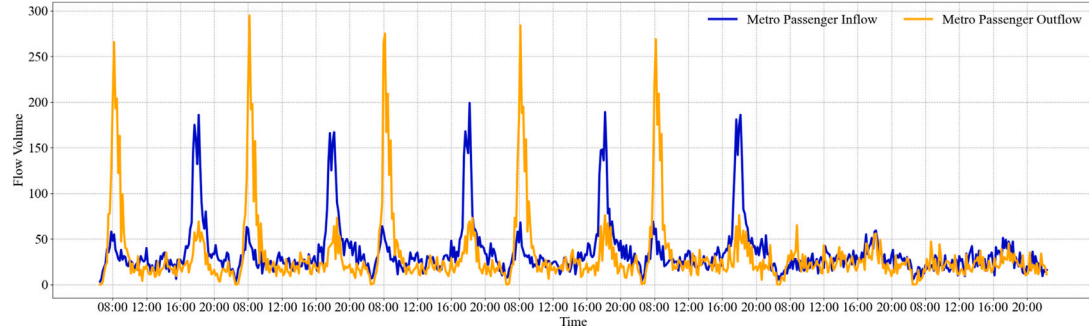


Fig. 2. Metro passenger flow in a week — Shuibu Station of Fuzhou Metro Line 2.

3. Methodology

In this section, the proposed method, which consists of three main parts, i.e. the two-step KMSFC, external feature fusion, and DVRGCN, is described in detail. The KMSFC algorithm and external feature fusion process, which gather the spatiotemporal information of the metro network, are significant for the effectiveness of the DVRGCN, which captures the spatial dependencies and temporal dynamics of the MPF data.

3.1. Notations and problem statement

A metro system network is represented as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the set of nodes representing the metro stations and $\mathcal{E}$ is a set of edges representing station connections. Let $e_{i,j}$ denote the edge between station v_i and v_j . We also set the number of metro stations to be N and the corresponding adjacency matrix to be $A \in \mathbb{R}^{N \times N}$. The observations of node v_i at time t are recorded as the vector $x_t^i \in \mathbb{R}^F$, where F is the dimension of the observation vector. If there are only the metro passenger inflow and outflow being observed at node v_i , its graph signal at time t is denoted as $x_t^i \in \mathbb{R}^2$. Thus, the observations of all metro stations at time t can be described by $X_t = [x_t^0, x_t^1, \dots, x_t^N] \in \mathbb{R}^{N \times F}$. The historical observation data in T time intervals is represented by $\mathcal{X}_T = [X_0, X_1, \dots, X_T] \in \mathbb{R}^{T \times N \times F}$.

The aim of this paper is to generate a function F_θ for forecasting precisely the metro passenger inflow and outflow of N metro stations for next T' time intervals, denoted by $\hat{\mathcal{X}}_{T'} \in \mathbb{R}^{T' \times N \times 2}$, based on the metro observation data of the past T time intervals $\mathcal{X}_T \in \mathbb{R}^{T \times N \times F}$ and the pre-constructed adjacency matrix A , as formulated in (1).

$$\hat{\mathcal{X}}_{T'} = F_\theta(\mathcal{X}_T; A). \quad (1)$$

3.2. Two-step KMSFC algorithm

The nature geographic adjacency graph of the metro system network only contains the simple neighborhood information, which is

insufficient to deliver the comprehensive graph information for the GCN to capture the spatial features. This paper proposes a novel two-step KMSFC algorithm to cluster the metro stations based on the dynamic MPF feature sequences. The dynamic MPF feature sequence proposed in this paper can explicitly explain the MPF changing patterns of the metro stations. In addition, the metro passenger inflow and outflow patterns are different, even in the same metro station. Therefore, the passenger travel behavior should be considered separately from those two perspectives. For convenience, the dynamic MPF feature sequence mentioned hereafter in this paper corresponds to either the passenger inflow or passenger outflow. A reminder is that the metro passenger flow patterns on special holidays differ significantly from those on normal weekdays and this paper concentrates on the metro passenger commuting patterns during the normal week, i.e. the week without a special holiday. A city can be divided into several functional areas such as the residential areas, workplaces, schools, and shopping centers. The metro passenger travel behavior analyses are critical for recognizing the spatial distribution of the urban functional areas (Sun & Axhausen, 2016). The daily MPF intuitively reflects the passenger travel behavior. For example, Fig. 1 shows the MPF of the metro station surrounded by the residential areas in one week. The metro passenger inflow peak time is usually around 8 am on weekdays, while the outflow peak hours start from 6 pm to 7 pm. On the weekends, both the inflow and outflow volumes are small. The MPF changing trends of the metro station clearly display the resident commuting behavior. On the contrary, the MPF variation of the station (as shown in Fig. 2) located around the workplace is totally different. The outflow volume in the morning rush hour rises significantly on weekdays, and the inflow volume jumps remarkably during the evening rush hour. By contrast, both the inflow and outflow volumes stay steady in low levels and there is no drastic change on the weekends.

The two examples above reveal that the MPF changing patterns of different metro stations are extremely different. Nevertheless, passenger travel patterns on weekdays/weekends are similar in these stations. In the case of metro stations connected to railway stations, the MPF variation patterns on weekdays and weekends are dissimilar. As shown

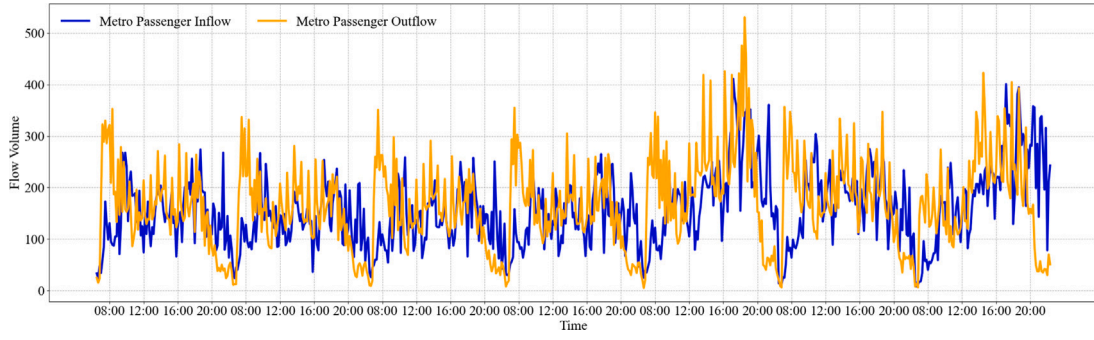


Fig. 3. Metro passenger flow in a week — Fuzhou railway station of Fuzhou Metro Line 1.

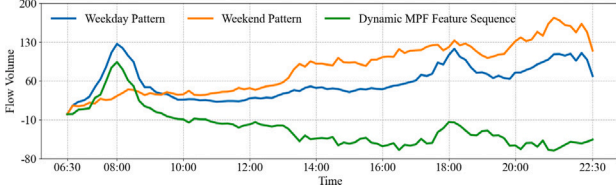


Fig. 4. Dynamic MPF sequence — Ninghua station of Fuzhou Metro Line 2.

in Fig. 3, the passenger travel patterns on Fridays and Sundays are similar because an amount of passengers get off (or get on) the train and arrive (or leave) on Friday and Sunday. For these stations, Fridays and Sundays should be consider as the weekend travel pattern and the rest days should be included in the weekday travel pattern. In different urban functional areas, the MPF evolution processes reveal different weekday and weekend patterns of passenger travel behavior. Simply dividing the MPF changing patterns in a week into the weekday and weekend patterns cannot fully demonstrate the passenger travel behavior in all metro stations. This paper adopts the K-means clustering algorithm to reclassify the “weekday” and “weekend” MPF patterns for all metro stations.

The historical MPF of metro station j is taken as an example to explain the two-step KMSFC algorithm. Suppose that there are m weeks and each day includes r time slots. Step one starts with calculating the average historical MPF sequences $[H^{1,j}, H^{2,j}, \dots, H^{7,j}]$ from Monday to Sunday. The average MPF $\bar{H}^{w,j}$ for each day in a week is calculated using (2). For each metro station, there are seven average MPF sequences for clustering and each sequence is considered as an r -dimensional data. The number of clusters is set as two, indicating the individual MPF patterns for weekdays and weekends, and the Euclidean distance is used as the distance measurement of sequences. The step-one clustering results contain the MPF sequences of new weekday and weekend patterns, which are represented by $H^{d,j}$ and $H^{e,j}$, respectively. The passenger commuting behavior in each metro station can be appropriately analyzed by the proposed dynamic MPF feature sequence, which is constructed by utilizing the difference between $H^{d,j}$ and $H^{e,j}$. First, the average weekday and weekend pattern sequences $\bar{H}^{d,j}$ and $\bar{H}^{e,j}$ are calculated by (3)–(4), where $n_{1,j}$ and $n_{2,j}$ are the number of sequences in the weekday and weekend pattern clusters, respectively. Next, the dynamic MPF feature sequence $\mathcal{P}^j$ is obtained in (5). An example of the inflow dynamic MPF feature sequence $\mathcal{P}^{sc}$ of a metro station located at a shopping center is shown as the green line in Fig. 4, where the blue and orange lines represent $H^{d,sc}$ and $H^{e,sc}$, respectively. There is a peak during the morning in the weekday pattern while the flow volume is low during the same time in the weekend pattern. The morning peak of the blue line illustrates that there are also some residential areas around the station so the commuting demand exists in the morning. Besides, the inflow volumes of both patterns increase constantly from 2 pm. As people living in cities usually knock

off around 6 pm and go shopping after work, the blue line shows a relative high inflow volume after 6 pm. The orange line rises from 1 pm and stays high in volume which explicitly displays that considerable people spend their time in the mall on weekends. The positive-value segment of the green line indicates the volume on weekdays being greater than that on weekends, and vice versa. The positive part of $\mathcal{P}^{sc}$ exists mainly in the morning period because the people living around the station go to work during the weekday morning. The rest are all negative, which explains that more people enter the station during the weekend afternoon than the weekday afternoon. The difference $\mathcal{P}^{sc}$ between the weekday and weekend patterns implies the feature of the passenger travel pattern of the metro station. In Step 2, the min-max normalization as shown in (6) is utilized for $\mathcal{P}^j$ to eliminate the influence of different magnitudes and reserve the pure variation trend of the MPF. Then, the second step of the algorithm is applied to all the dynamic MPF feature sequences for all the metro stations. All settings are the same as the first step expect the number of clusters being K . After the two-step K-means clustering, K groups of the metro stations with similar dynamic MPF features are obtained. The full two-step clustering process is demonstrated in Algorithm 1.

$$\begin{aligned} \bar{H}^{w,j} &= \frac{1}{m} \sum_{i=1}^m H_i^{w,j} \\ &= \left[\frac{1}{m} \sum_{i=1}^m h_{i,0}^{w,j}, \frac{1}{m} \sum_{i=1}^m h_{i,1}^{w,j}, \dots, \frac{1}{m} \sum_{i=1}^m h_{i,r}^{w,j} \right], \\ w &\in \{1, \dots, 7\}, \end{aligned} \quad (2)$$

$$\begin{aligned} \bar{H}^{d,j} &= \frac{1}{n_{1,j}} \sum_{i=1}^{n_{1,j}} H_i^{d,j} \\ &= \left[\frac{1}{n_{1,j}} \sum_{i=1}^{n_{1,j}} h_{i,0}^{d,j}, \frac{1}{n_{1,j}} \sum_{i=1}^{n_{1,j}} h_{i,1}^{d,j}, \dots, \frac{1}{n_{1,j}} \sum_{i=1}^{n_{1,j}} h_{i,r}^{d,j} \right], \end{aligned} \quad (3)$$

$$\begin{aligned} \bar{H}^{e,j} &= \frac{1}{n_{2,j}} \sum_{i=1}^{n_{2,j}} H_i^{e,j} \\ &= \left[\frac{1}{n_{2,j}} \sum_{i=1}^{n_{2,j}} h_{i,0}^{e,j}, \frac{1}{n_{2,j}} \sum_{i=1}^{n_{2,j}} h_{i,1}^{e,j}, \dots, \frac{1}{n_{2,j}} \sum_{i=1}^{n_{2,j}} h_{i,r}^{e,j} \right], \end{aligned} \quad (4)$$

$$\mathcal{P}^j = \bar{H}^{d,j} - \bar{H}^{e,j} = [p^{0,j}, p^{1,j}, \dots, p^{r,j}], \quad (5)$$

$$p^{*,j} = \frac{p^{*,j} - p_{min}^j}{p_{max}^j - p_{min}^j}. \quad (6)$$

3.3. External features

To achieve a precise MPF prediction, this study considers the external features, including day of the week, time of the day, and volume level of the MPF. The goal of utilizing the external features is to make the DL model identify the passenger patterns in different time periods as well as the stations with different MPF volume levels. In this section,

Algorithm 1 Two-Step K-means-based Metro Station Functional Clustering

1: INPUT: History m weeks inflow/outflow of N metro stations

$$H = [H^1, H^2, \dots, H^m] \in \mathbb{R}^{m \times 7 \times N \times r}$$

2: OUTPUT: K clusters of metro stations

$$\bar{H}^w \in \mathbb{R}^{7 \times N \times r} \leftarrow \frac{1}{m} \sum_{i=1}^m H_i^w, w \in \{1, \dots, 7\}$$

3: FOR: Metro station $j, j \in \{1, 2, \dots, N\}$ DO :

$$4: \bar{H}^{w,j} = \frac{1}{m} \sum_{i=1}^m H_i^{w,j} \in \mathbb{R}^{7 \times r}$$

$$= \left[\frac{1}{m} \sum_{i=1}^m h_{i,0}^{w,j}, \frac{1}{m} \sum_{i=1}^m h_{i,1}^{w,j}, \dots, \frac{1}{m} \sum_{i=1}^m h_{i,r}^{w,j} \right]$$

$$5: H^{d,j} \in \mathbb{R}^{n_{1,j} \times r}, H^{e,j} \in \mathbb{R}^{n_{2,j} \times r} \leftarrow \text{K-means}(\text{clusters} = 2)(\bar{H}^{w,j})$$

$$6: \bar{H}^{d,j} = \frac{1}{n_{1,j}} \sum_{i=1}^{n_{1,j}} H_i^{d,j} \in \mathbb{R}^{1 \times r}$$

$$= \left[\frac{1}{n_{1,j}} \sum_{i=1}^{n_{1,j}} h_{i,0}^{d,j}, \frac{1}{n_{1,j}} \sum_{i=1}^{n_{1,j}} h_{i,1}^{d,j}, \dots, \frac{1}{n_{1,j}} \sum_{i=1}^{n_{1,j}} h_{i,r}^{d,j} \right]$$

$$7: \bar{H}^{e,j} = \frac{1}{n_{2,j}} \sum_{i=1}^{n_{2,j}} H_i^{e,j} \in \mathbb{R}^{1 \times r}$$

$$= \left[\frac{1}{n_{2,j}} \sum_{i=1}^{n_{2,j}} h_{i,0}^{e,j}, \frac{1}{n_{2,j}} \sum_{i=1}^{n_{2,j}} h_{i,1}^{e,j}, \dots, \frac{1}{n_{2,j}} \sum_{i=1}^{n_{2,j}} h_{i,r}^{e,j} \right]$$

$$8: p^j = \bar{H}^{d,j} - \bar{H}^{e,j} = [p^{0,j}, p^{1,j}, \dots, p^{r,j}] \in \mathbb{R}^{1 \times r}$$

$$9: \text{Min-Max Normalization: } p^{*,j} = \frac{p^{*,j} - p_{\min}^j}{p_{\max}^j - p_{\min}^j}$$

10: ENDFOR

$$11: P = [p^1, p^2, \dots, p^N] \in \mathbb{R}^{N \times r}$$

12: Metro Station Clusters $\leftarrow \text{K-means}(\text{clusters} = K)(P)$

the acquisition and one-hot representations of the external features will be introduced in detail. The time-related features could have significant influences on the final forecasting accuracy. The daily evolution pattern of the MPF is highly similar. For instances, the MPFs of each Tuesday in five weeks are shown in Fig. 5. The day of week indicates the daily MPF evolution pattern and the one-hot representation is $\alpha_{day} = [0, 1, 0, 0, 0, 0, 0] \in \mathbb{R}^{1 \times 7}$. Furthermore, the time of day as a more refined time feature is considered as well. The representation of the time of day is a value ranging from zero to one. If the time intervals of MPF in one day is set as r , the value of q th time interval is $\frac{q}{r}$. The final one-hot representation of time feature at q th time interval in a day for the metro station is $\alpha'_q = \left[\alpha_{day}, \frac{q}{r} \right] \in \mathbb{R}^{1 \times 8}$, where $q \in [0, r)$. Finally, the indicator of time $\alpha_q \in \mathbb{R}^{N \times 8}$ for N metro stations is the N -time duplication of α'_q . Another external feature is the MPF volume level of each metro station. As shown in Fig. 6, metro stations have tremendous differences in throughput due to the capacity and location. Metro stations located in high density population areas such as shopping mall, railway stations and transfer stations have larger passenger flows. Passenger travel behavior patterns in those stations are more complex. Therefore, this paper devises a novel metro station volume level one hot representation method to present MPF volume feature for each metro station. The MPF volume levels of metro stations represent the overall MPF volume differences among all stations. For simplicity, the MPF indicate either passenger inflow or outflow. The general volume level one hot representation of metro station i is $\beta_i = [b_1, b_2, \dots, b_i, \dots, b_{2(c+1)}] \in \mathbb{R}^{1 \times 2(c+1)}$, where $2(c+1)$ is the max volume level. The first step is to calculate mean μ and standard deviation σ of N metro stations MPF, which are shown in (7)–(8). If $MPF^i < \mu - \eta\sigma$, the encoding of i th station is $\beta_i = [b_1, \dots, b_{c+1-\eta}, 0, \dots, 0]$. In β_i , $c+1-\eta > 0$ and values from b_1 to $b_{c+1-\eta}$ are 1. When $MPF^i > \mu + \eta\sigma$, values of β_i from b_1 to $b_{c+2+\eta}$ are set 1 where $c+2+\eta < 2(c+1)$ and the rest bits are 0. Since there are N metro stations, the final volume level one hot encoding is $\beta \in \mathbb{R}^{N \times 2(c+1)}$. Then, MPF volumes of all metro stations are compressed into a one-hot representation. Ultimately, external features will be embedded into the proposed DVRGCN through fully-connected layers.

$$\mu = \frac{1}{N} \sum_{i=1}^N MPF^i. \quad (7)$$

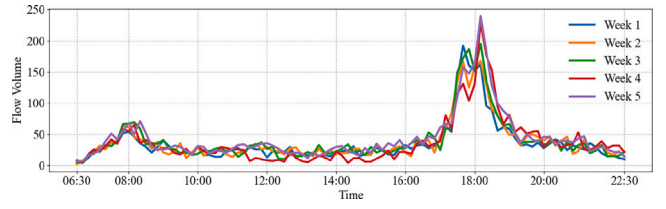


Fig. 5. Metro passenger inflow on Tuesdays in five weeks — Shuibu station of Fuzhou Metro Line 2.

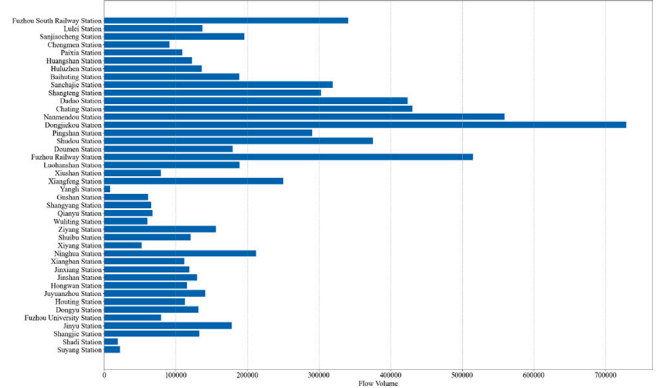


Fig. 6. Statistics of metro passenger inflow in five weeks — Fuzhou Metro System.

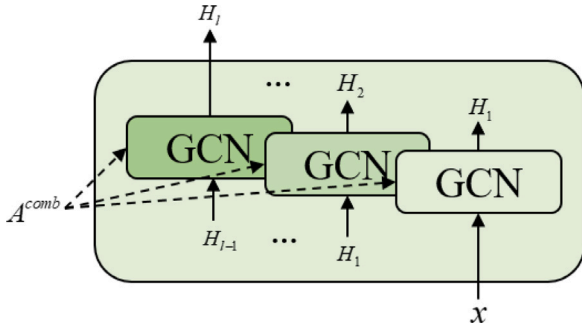
$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (MPF^i - \mu)^2}. \quad (8)$$

3.4. The proposed metro passenger flow prediction model

In this section, the DVRGCN is formulated to learn the spatiotemporal features of the MPF data for achieving precise MPF predictions. The GCN in spectral domain and self-adaptive GCN are from two different views, which are the local and hidden spatial feature learning of the metro traffic network. The adjacency matrix for the GCN and the construction process for the multiple GCN layers are demonstrated in Section 3.4.1. The self-adaptive GCN which serves as a supplemental module to capture hidden spatial relations over the metro traffic network is introduced in Section 3.4.2. Finally, Section 3.4.3 gives the details of the proposed DVRGCN model.

3.4.1. Local spatial feature learning

Graph Construction. The graph with prior knowledge of the practical metro traffic network is important for the GCN module to effectively capture the spatial features. Previous studies simply considered static information (Chen et al., 2022; Liu et al., 2022) in constructing the graphs for the GCN modules extracting the spatial features. However, those graphs fail to include the semantic information of the metro network. In this study, two types of graphs with wealth information are combined to make a comprehensive graph. The first graph is constructed based on the KMFSC results since the metro stations in the same cluster have similar MPF changing trends and are located at similar urban functional areas. In the graph, those metro stations in the same cluster will be assigned virtual connections. Based on the graph, $A^{cluster}$ reflects the latent urban functional dependencies and hidden relations of the metro traffic network. Specifically, the element $A_{i,j}^{cluster}$ is calculated in (9). The second graph is a ξ -minute arrival graph, and the connection between two stations is based on the train arrival time. It takes ξ minutes to travel from station i to station j , and there are several stations between i and j . In ξ minutes, passengers get on the

Fig. 7. The architecture of l -layer MGCN.

train at the stations between i and $j - 1$ and some passengers get off the train at the stations between $i + 1$ and j . The short-term MPF variations of these stations are highly related. An arrival matrix $A^{arrival}$ originates from the arrival graph. If the time δ of the train traveling from one station to another is less than ξ minutes, these two metro stations are connected. The specific element in $A^{arrival}$ is demonstrated in (10). The final comprehensive matrix A^{comb} includes both the urban functional and practical geographical information. The calculation of A^{comb} is demonstrated in (11).

$$A_{i,j}^{cluster} = \begin{cases} 1, & \text{if } i, j \text{ in the same cluster and } i \neq j \\ 0, & \text{otherwise} \end{cases}, \quad (9)$$

$$A_{i,j}^{arrival} = \begin{cases} 1, & \text{if } \delta < \xi \\ 0, & \text{otherwise} \end{cases}, \quad (10)$$

$$A^{comb} = A^{cluster} + A^{arrival}. \quad (11)$$

Multi-layer GCN Module. The GCN introduced in this section is from the perspective of spectral domain with the Fourier transformation (Bruna et al., 2013). Let $A \in \mathbb{R}^{N \times N}$ be the adjacency matrix and D be the diagonal degree matrix of A . The random walk Laplacian is $L^{rw} = D^{-1}L = I_N - D^{-1}A$, where $I_N \in \mathbb{R}^{N \times N}$ is the identity matrix. Through the eigen-decomposition $L^{rw} = P\Lambda P^T$, the eigenvectors $P \in \mathbb{R}^{N \times N}$ and the corresponding eigenvalues $\Lambda \in \mathbb{R}^{N \times N}$ are obtained. Suppose that the graph signal is $x \in \mathbb{R}^{N \times F}$ and the Fourier transformation is $P^T x$. The graph convolution is represented in (12), where $g_\theta(\Lambda)$ is the learnable convolution kernel. To accelerate the computation of graph convolution process, the Chebyshev polynomial approximation (Defferrard, Bresson, & Vandergheynst, 2016) is adopted to reduce the number of parameters. In (13), $g_\theta(\Lambda)$ is approximated by the Chebyshev polynomial of order k , i.e. $T_k(\tilde{\Lambda})$, where θ_k is the trainable parameter and λ_{max} is the max eigenvalue in Λ . The recursive formula of $T_k(x)$ is shown in (14), and the final graph convolution formula is transformed into (15). A multi-layer GCN (MGCN) module shown in Fig. 7 is proposed to extract the local spatial features of the metro traffic network. As mentioned in Section 3.1, the metro observation data of N nodes at the historical time t is $X_t \in \mathbb{R}^{N \times F}$. The convolution result of l th layer shown in (16) is $H_l \in \mathbb{R}^{N \times F'}$, where $H_0 = X_t$, F' is the output dimensions of the l th layer and $Dropout(\cdot)$ denotes the dropout function.

$$(x * g_\theta)_{GCN} = g_\theta(P\Lambda P^T)x = P g_\theta(\Lambda) P^T x, \quad (12)$$

$$g_\theta(\Lambda) \approx \sum_{k=0}^{K-1} \theta_k T_k(\tilde{\Lambda}), \text{ where } \tilde{\Lambda} = 2\Lambda/\lambda_{max} - I_N, \quad (13)$$

$$T_k(x) = 2xT_{k-1}(x) - T_{k-2}(x), \quad (14)$$

$$\text{where } T_0(x) = 1, T_1(x) = x, \\ (x * g_\theta)_{GCN} = \sum_{k=0}^{K-1} \theta_k T_k(\tilde{L})x, \text{ where } \tilde{L} = P\tilde{\Lambda}P^T, \quad (15)$$

$$H_l = Dropout((H_{l-1} * g_\theta)_{GCN}) \\ = Dropout\left(\sum_{k=0}^{K-1} \theta_k T_k(\tilde{L})H_{l-1}\right). \quad (16)$$

3.4.2. Self-adaptive hidden spatial feature learning

The traditional GCN highly relies on the empirical Laplace matrix of the traffic graph, which is not capable of capturing the complete spatial features of the entire metro traffic network. Recent researches (Guo et al., 2022b; Ta et al., 2022) adopted the self-adaptive adjacency matrix to automatically learn hidden spatial features of the traffic network. The self-adaptive adjacency matrix $A^{adaptive}$ is the multiplication of the two trainable vectors $E_1 \in \mathbb{R}^{N \times Q}$ and $E_2 \in \mathbb{R}^{N \times Q}$. The specific construction formula of $A^{adaptive}$ is shown in (17). Those metro stations with weak correlations contribute to small even negative values in $A^{adaptive}$. The activation function $ReLU$ is adopted as a filter. The self-adaptive adjacency matrix $A^{adaptive}$ is not symmetric, and this paper adopts the diffusion convolution (Wu et al., 2019) on the graph to adaptively capture the spatial features of the metro traffic network. Then, the adaptive GCN (AGCN) module is applied on the metro traffic network to achieve the hidden spatial feature learning. The metro observation data at the historical time t is $X_t \in \mathbb{R}^{N \times F}$, and the convolution layer finally outputs the K -step diffusion convolution result $Z \in \mathbb{R}^{N \times F'}$ in (18).

$$A^{adaptive} = ReLU(E_1 E_2^T), \quad (17)$$

$$Z = \sum_{k=0}^K (A^{adaptive})^k X_t W_k. \quad (18)$$

3.4.3. Dual-view recurrent graph convolution network

The Seq2Seq architecture of the proposed DVRGCN is shown in Fig. 8. Based on the architecture, the proposed DVRGCN is capable of encoding the spatial and temporal features simultaneously. Then, the precise prediction is achieved through the decoder. The core of the DVRGCN is the gated recurrent spatial unit (GRSU), and the number of the GRSU layers is γ . The GRSU contains the dual-view GCN modules which capture both the explicit and hidden spatial features of the metro traffic network. Moreover, the external features are embedded into the DVRGCN to capture the patterns of critical time periods and different MPF volume levels.

Overview. Fig. 8 displays an overview of data flow in the proposed DVRGCN model. The encoder of the DVRGCN inputs the MPF observation data with T historical time intervals $\mathcal{X}_T = [X_0, X_1, \dots, X_T] \in \mathbb{R}^{T \times N \times F}$. With the concatenation of external features α and β , the encoder learns both spatiotemporal patterns and external features through the GRSU along the time axis. Then, the decoder of the DVRGCN recurrently fuses the captured metro traffic patterns and “key” feature C_f which is compound of α and β . Finally, the proposed model predicts the metro passenger inflow and outflow for the future T' time intervals $\mathcal{Y}_{T'} = [Y_{T+1}, Y_{T+2}, \dots, Y_{T+T'}] \in \mathbb{R}^{T' \times N \times 2}$.

External Features Fusion. The external features considered in this paper include time features α and the MPF volume level β . To fully obtain the influences of external features, α and β are embedded into the DVRGCN through fully-connected layers. In the encoder part of the DVRGCN, α of the MPF is important for the DVRGCN to recognize the MPF changing patterns during fluctuated periods. As shown in the left side of Fig. 8, α at each time step is embedded into the hidden state and the GRSU will learn the implicit time feature of the MPF. Let one hot representation of time at historical time t be $\alpha_t \in \mathbb{R}^{N \times 8}$. Then, α_t is fed into the full connected layer $FC(\cdot)$ and the output will be concatenated with the hidden state h_t to form the new hidden state h'_t for the GRSU. The complete fusion process is illustrated in (19). The MPF volume level β denotes the inflow and outflow volume level. The final MPF volume level feature is $\beta \in \mathbb{R}^{N \times 2 \times 2(c+1)}$ which is concatenated by the metro observation data $X_t \in \mathbb{R}^{N \times 2 \times 2(c+1)}$ at the historical time t . Thus, the MPF volume level patterns of all the metro stations will be learned by the GRSU when capturing the spatial features of the MPF data. With the incorporation of external features, the DVRGCN effectively encodes the spatiotemporal features and external features. Prior to the final decoder output (as shown in the right side of Fig. 8),

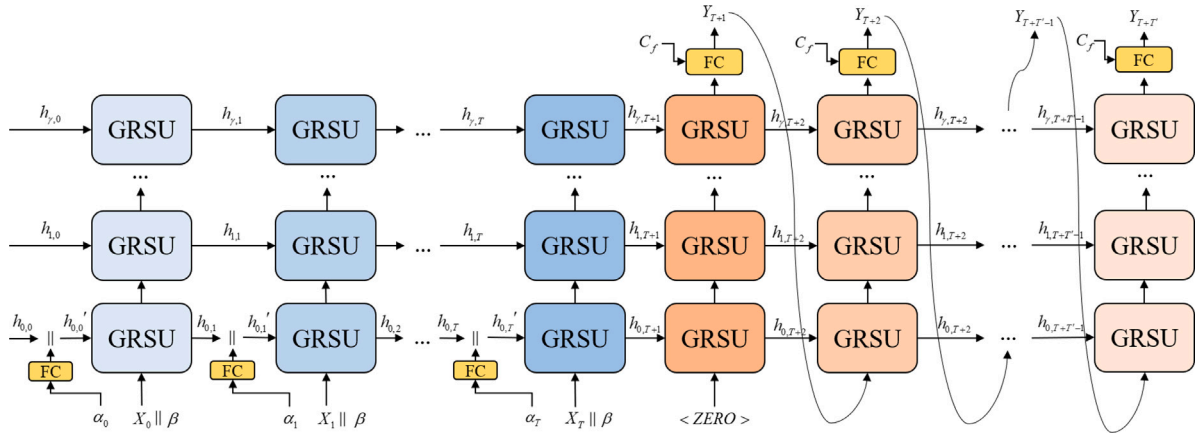


Fig. 8. The architecture of the proposed DVRGCN.

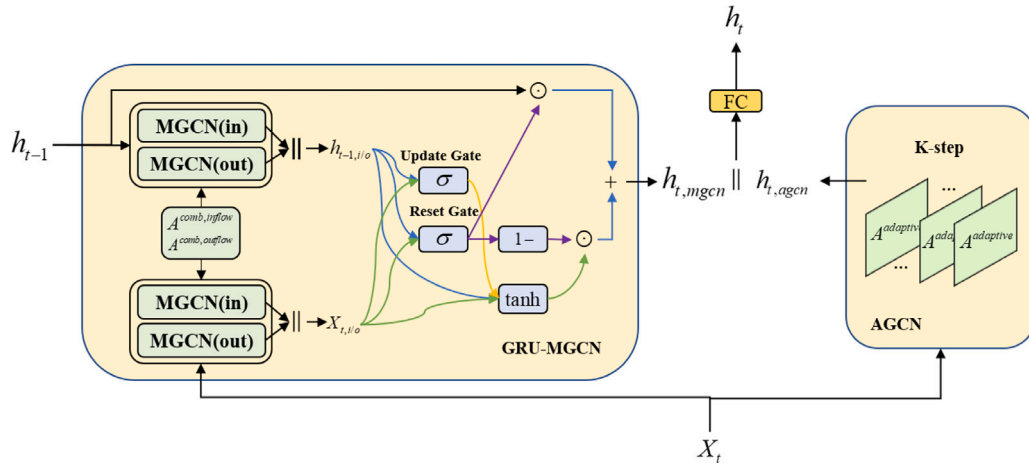


Fig. 9. The architecture of the proposed DVRGCN.

the compound external factor C_f concatenating both $\alpha \in \mathbb{R}^{N \times 8 \times T}$ and $\beta \in \mathbb{R}^{N \times 2 \times (c+1)}$ is transformed as the “key” for decoding the output the GRSU. The transformation is made through a fully-connected layer which is shown in (20).

$$h'_t = FC(\alpha_t) \parallel h_t. \quad (19)$$

$$C_f = FC(\alpha \parallel \beta). \quad (20)$$

Gated Recurrent Spatial Unit. As the core of DVRGCN, the GRSU consists of two components: GRU-MGCN and AGCN. The dual views of the GCN ensure the maximum preservation of spatial information of the metro traffic network. The GRU is effective in the temporal feature extraction while the operations in the GRU are usually performed with full connection. The GRU-MGCN proposed in this paper replaces the full connection of the GRU with the MGCN to achieve the spatial feature extraction. The structure of the GRU-MGCN is shown in the left side of Fig. 9. The GRU-MGCN is used to determine the important local spatial features along the time axis to be memorized by the model. The metro passenger inflow and outflow patterns are different even in the same station. Therefore, this paper adopts two MGCN modules to capture the inflow and outflow spatial features respectively. Thus, there are two adjacency graphs $A^{comb,inflow}$ and $A^{comb,outflow}$ for the MGCN modules. Let $X_t \in \mathbb{R}^{N \times F}$ be the input at time t and the hidden state from $t-1$ be h_{t-1} . First, X_t is divided equally into $X_{t,in} \in \mathbb{R}^{N \times (F/2)}$ and $X_{t,out} \in \mathbb{R}^{N \times (F/2)}$, which represent the observations of the metro passenger inflow and outflow, respectively. Then $X_{t,in}$ and $X_{t,out}$ are fed into two MGCNs to capture local spatial features separately and the outputs

of two MGCNs are concatenated as $X_{t,i/o}$. The GRU-based recurrent structure captures the temporal patterns of $X_{t,i/o}$. The same operation is applied to h_{t-1} , and the concatenated output is represented by $h_{t-1,i/o}$. The graph convolution of the MGCN is abbreviated by $x * g_\theta$. Let $u_{t,i/o}$, $r_{t,i/o}$, and $n_{t,i/o}$ denote the update gate, reset gate, and extracted spatiotemporal features. The specific calculations are displayed in (21)–(23), where $\odot$ denotes the element-wise multiplication and $\parallel$ indicates the concatenation operation. Let $h_{t,mgcn}$ be the current output of the GRU-MGCN, which selectively outputs the valuable information of $n_{t,i/o}$ and $h_{t-1,i/o}$. The calculation of $h_{t,mgcn}$ is shown in (24). The hidden spatial features are captured by the AGCN in the right side of Fig. 9. The input X_t of the AGCN is the same as that of the GRU-MGCN, and the output is $h_{t,agcn}$ shown in (25). There is a fully-connected layer $FC(\cdot)$ to transfer the fusion of $h_{t,mgcn}$ and $h_{t,agcn}$ into h_t . The final output h_t can be represented in (26).

$$u_{t,i/o} = \sigma(X_{t,i/o} * g_{Xu,i/o} + h_{t-1,i/o} * g_{hu,i/o} + b_{u,i/o}), \quad (21)$$

$$r_{t,i/o} = \sigma(X_{t,i/o} * g_{Xr,i/o} + h_{t-1,i/o} * g_{hr,i/o} + b_{r,i/o}), \quad (22)$$

Through the operations in the GRSU, the DVRGCN effectively captures spatiotemporal features of the metro traffic network.

$$n_{t,i/o} = \tanh(X_{t,i/o} * g_{Xn,i/o} + u_{t,i/o} \odot h_{t-1,i/o} * g_{hn,i/o} + b_{n,i/o}) \quad (23)$$

$$h_{t,mgcn} = (1 - r_{t,i/o}) \odot n_{t,i/o} + r_{t,i/o} \odot h_{t-1,i/o}, \quad (24)$$



Fig. 10. The map of Fuzhou metro.

$$h_{t,agcn} = \sum_{k=0}^{K_2} (A^{adaptive})^k X_t W_k, \quad (25)$$

where $A^{adaptive} = ReLU(E_1 E_2^T)$,

$$h_t = Dropout(FC(h_{t,mgcn} \parallel h_{t,agcn})). \quad (26)$$

3.4.4. Loss function

The metro traffic observation data $\mathcal{X}_T = [X_0, X_1, \dots, X_T] \in \mathbb{R}^{T \times N \times F}$ is fed into the DVRGCN for training. The DVRGCN recursively generates the prediction through T' steps, shown as $\mathcal{Y}_{T'} = [Y_{T+1}, Y_{T+2}, \dots, Y_{T+T'}] \in \mathbb{R}^{T' \times N \times 2}$. The mean absolute error (MAE) is chosen as the loss function. Suppose that the ground truth of the MPF is $\hat{\mathcal{Y}}_{T'} = [\hat{Y}_{T+1}, \hat{Y}_{T+2}, \dots, \hat{Y}_{T+T'}] \in \mathbb{R}^{T' \times N \times 2}$, and the loss function is defined by (27).

$$\begin{aligned} Loss(\mathcal{Y}_{(T+1):(T+T')}, \hat{\mathcal{Y}}_{(T+1):(T+T')}) \\ = \frac{1}{T' * N * 2} \sum_{i=1}^{T'} \sum_{j=1}^N \sum_{k=1}^2 \|Y_{(T+i),j,k} - \hat{Y}_{(T+i),j,k}\|. \end{aligned} \quad (27)$$

4. Experiments

In this section, a practical metro traffic dataset is utilized on the experiments for evaluating the proposed model by comparison with the state-of-the-art methods. Three evaluation metrics are adopted to comprehensively demonstrate the method performance in statistics. In addition, the ablation experiments are used to illustrate the contributions of individual modules in the proposed DVRGCN.

4.1. Dataset

The practical MPF data is collected by the AFC system of Fuzhou Metro. There are two lines in Fuzhou Metro (shown in Fig. 10) which are Line one and Line two. The raw data includes data fields: *LineID*, *TransactionID*, *TransactionTime*, *EntryStation*, *ExitStation*, *EntryTime*, *ExitTime*, which are used to count the statistics of the metro passenger inflow and outflow. The time ranges from May 6th 2019 to June

30th 2019. There is one week containing a special holiday and the passenger behavior is greatly different from that in normal weeks. Therefore, the MPF data of that week is eliminated from the original dataset. The final MPF dataset including 42 metro stations records the statistics of inflow and outflow in seven weeks. The metro operation time starts from 6:30 am to 10:30 pm and the time interval of the MPF data is set to be 10 min. Thus, the MPF data in one day consists of 96 intervals. The experiments adopt the MPF observation data of six historical time intervals (i.e. 60 min) to predict the metro passenger inflow and outflow for six future time intervals (i.e. 60 min). The Z-score normalization in (28) is applied to the metro observation data $\mathcal{X}$, where $\mathcal{X}'$ is the normalized data. Let $\mu_{\mathcal{X}}$ and $\sigma_{\mathcal{X}}$ be the mean and standard deviation of $\mathcal{X}$, respectively. Since there are seven weeks, the ratio of 5:1:1 is taken to separate the training set, validation set, and test set. Seven-fold cross validation is adopted to get the stable performance for all methods.

$$\mathcal{X}' = \frac{\mathcal{X} - \mu_{\mathcal{X}}}{\sigma_{\mathcal{X}}} \quad (28)$$

4.2. Evaluation metrics

Used for the experiments are three evaluation metrics, including (1) mean absolute error (MAE), (2) root mean squared error (RMSE), and (3) mean absolute percentage error (MAPE). Then, the evaluation metrics are defined in (29)–(31), where $\mathcal{N}$ is the number of the MPF samples, $\mathcal{Y} = [Y_1, Y_2, \dots, Y_{\mathcal{N}}]$ represents the prediction, and $\hat{\mathcal{Y}} = [\hat{Y}_1, \hat{Y}_2, \dots, \hat{Y}_{\mathcal{N}}]$ is the ground truth.

$$MAE(\mathcal{Y}, \hat{\mathcal{Y}}) = \frac{1}{\mathcal{N}} \sum_{i=1}^{\mathcal{N}} \|Y_i - \hat{Y}_i\|, \quad (29)$$

$$MAPE(\mathcal{Y}, \hat{\mathcal{Y}}) = \frac{1}{\mathcal{N}} \sum_{i=1}^{\mathcal{N}} \left\| \frac{Y_i - \hat{Y}_i}{\hat{Y}_i} \right\| \times 100\%, \quad (30)$$

$$RMSE(\mathcal{Y}, \hat{\mathcal{Y}}) = \sqrt{\frac{1}{\mathcal{N}} \sum_{i=1}^{\mathcal{N}} (Y_i - \hat{Y}_i)^2}, \quad (31)$$

4.3. Baseline approaches

The proposed DVRGCN is compared with the following baseline methods:

1. History average (HA). The HA is a naive statistics method. In the experiment, the MPF at next time interval is the average of the MPF in past six time intervals. Since experiments require forecasting MPFs in next six time intervals, the predicted MPFs are also considered as the historical data for predicting MPF in next time interval.
2. Multilayer perceptron (MLP). The MLP conducted on experiments contains three fully-connected layers. The sizes of the input and output layers are both $2 * N * T'$ and the hidden layer consists of 256 neurons.
3. Long short-term memory (LSTM). This RNN-based model adopts two full-connected layers: the LSTM with hidden size 256 and a full-connected output layer.
4. Gated recurrent unit (GRU). This RNN-based model adopts two full-connected layers: the GRU with hidden size 256 and a full-connected output layer.
5. Graph convolution network and stacked bidirectional unidirectional long short-term memory network (GCN-SBULSTM) (Chen et al., 2022). The network contains a LSTM layer and a GCN layer in parallel to capture the spatiotemporal features of the metro traffic data and forecast the future MPF. Except the time span being set as six, we follow the reference paper for other hyperparameter settings of the GCN-SBULSTM.
6. Diffusion convolutional recurrent neural network (DCRNN) (Li et al., 2017). The Seq2Seq model with the diffusion convolution process extracts both the spatial and temporal information of traffic data. Except the time span being set as six, we again follow the reference paper for other hyperparameter settings of the DCRNN.
7. Physical-virtual collaboration graph network (PVCN) (Liu et al., 2022). The Seq2Seq model is coupled with the GCN-based recurrent unit to predict the future metro passenger inflow and outflow. Except the time span being set as six, the reference paper is followed for other hyperparameter settings of the PVCN.
8. Graph waveNet (Wu et al., 2019). The network includes the graph convolution layer and temporal convolution layer to capture the spatiotemporal features of the traffic network. Except the time span being set as six, other hyperparameter settings of the graph waveNet are done by following the reference paper.
9. Adaptive spatiotemporal graph neural Network (Ada-STNet) (Ta et al., 2022). The model includes the adaptive graph convolution layer and temporal convolution layer. The two-stage training helps capture both micro-level and macro-level spatial features of the traffic network. Except the time span being set as six, we follow the reference paper for other hyperparameter settings of the Ada-STNet.

The architecture comparisons between each baseline method and the proposed model are shown in Table 1. The traditional HA method only takes the historical data and calculates the average as the prediction. The MLP simply learns the nonlinear features of traffic data. Thus, both the HA and MLP fail to consider the spatiotemporal features of traffic data. The LSTM and GRU are the RNN variants which consider the temporal features of traffic data. Recently, the DL-based traffic prediction methods include the complex spatial or temporal modules for effectively capturing the spatiotemporal features of traffic data. The third row of Table 1 shows that only the DVRGCN considers the external features. In this study, the DCRNN, PVCN, and DVRGCN adopt the Seq2Seq architecture to recurrently learn the temporal features. The Ada-STNet, graph waveNet, and GCN-SBULSTM construct the independent temporal modules to extract the temporal patterns of traffic data. The spatial features of traffic data are extremely important

in traffic prediction. Recently, the GCN outperforms other approaches in the spatial feature learning of the traffic network and achieves high accuracy in traffic prediction. As shown in Table 1, those methods considering the spatial features all contain the GCN modules. There are two types of the GCN which utilize the self-adaptive and static matrices. The self-adaptive graph is constructed by the learnable parameters, and the elements of the matrix is updated automatically during the training process. The pre-defined graph is static and contains the prior knowledge of the traffic network. The Ada-STNet, Graph WaveNet, and DVRGCN adopt both adaptive and predefined adjacency matrices to fully capture the spatial features of traffic data. The self-adaptive graph is abandoned by the DCRNN, PVCN and GCN-SBULSTM.

4.4. Experimental settings

The parameter settings of the proposed DVRGCN are illustrated in this section. In the Seq2Seq architecture, there are two GRSU layers. Inside the GRSU, there are two components GRU-MGCN and AGCN. Variation patterns of the metro passenger inflow and outflow have great differences. Thus, there are two MGCN modules for capturing the spatial features of metro passenger inflow and outflow. The MGCN contains a two-layer GCN, and the output size in (16) is 128. The time interval in this experiment is set as 10 min. Besides, ξ of $A_{arrival}$ is also set to be 10 which includes the short-term information of the train passing stations in 10 min. The cluster number K of the KMSFC is set as eight. In the AGCN, there are three steps of diffusion convolution and the output size is 256. In (17), the vector size is $N \times 10$. The external features are transformed from the original dimensions into 256 through the fully-connected layers. The proposed model is trained by the Adam optimizer with an initial learning rate of 0.001. The training epoch is set to be 100, and the learning rate will reduce to 0.1 time the previous one at 70 and 90 epoch. The dropout rate is set as 0.3. To accelerate the convergence of the Seq2Seq model training, the training process adopts the curriculum learning (Bengio, Louradour, Collobert, & Weston, 2009). At the beginning stage of the training process, the input for the decoder at the next time step is the true label with the high probability. With the increase of training epochs, the probability goes down. In the later training stage, the decoder with a high probability uses the output from the last output which avoids the overfitting problem. All codes are implemented by PyTorch. The experiments are conducted under a computer equipped with Intel(R) Core(TM) i7-10700 2.90 GHz CPU and NVIDIA RTX 3070 GPU.

4.5. Experiments and analysis

The details of the experiments are illustrated in this section. Section 4.5.1 shows the experimental results for different cluster numbers of the K sets. In addition, the clustering results and the practicability of the KMSFC are analyzed. The MPF prediction performance of each compared method on the practical MPF data of Fuzhou metro is demonstrated in Section 4.5.2.

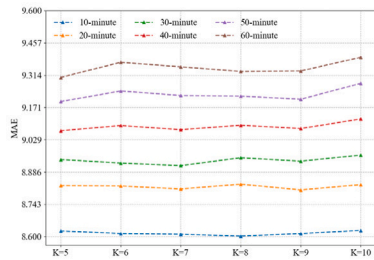
4.5.1. Metro station functional clustering

The number of cluster number K is manually defined. The experiments are conducted on different values of K to find the best number of clusters. The KMSFC takes around 0.34 s to complete the whole process for clustering the passenger flow data from Fuzhou Metro. Figs. 11–12 show the prediction errors when K equals 5, 6, 7, 8, 9 or 10. The value $K = 5$ makes the relatively low MAEs and RMSEs in the metro inflow predictions when the horizons are from 30 to 60 min. However, the prediction errors of other horizons are high when $K = 5$. The experiments of $K = 6, 7, 8, 9$ have similar performances on both inflow and outflow predictions. For the 10-minute horizon, $K = 8$ achieves best performance in inflow prediction. While in outflow prediction, $K = 6$ and $K = 8$ produce similar MAEs and RMSEs and $K = 8$ has the lowest MAPE. $K = 5, 6, 7, 8$ obtain similar MAEs and RMSEs in

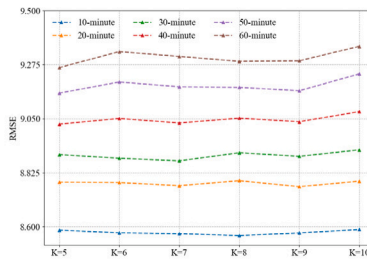
Table 1

Architecture comparisons between each baseline method and DVRGCN.

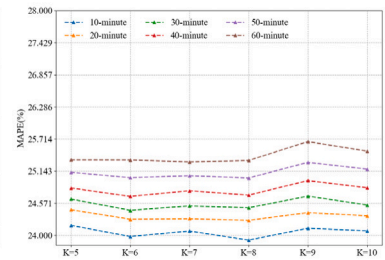
Differences	HA	MLP	LSTM	GRU	GCN-SBULSTM	DCRNN	PVGCN	Graph WaveNet	Ada-STNet	DVRGCN
Spatial Features	×	×	×	×	√	√	√	√	√	√
Temporal Features	×	×	√	√	√	√	√	√	√	√
External Features	×	×	×	×	×	×	×	×	×	√
Seq2Seq Architecture	×	×	×	×	×	√	√	×	×	√
Independent Temporal Feature Extraction Module	×	×	√	√	√	×	×	√	√	×
GCN (Self-adaptive Adjacency Matrix)	×	×	×	×	×	×	×	√	√	√
GCN (Static Matrix)	×	×	×	×	√	√	√	√	√	√



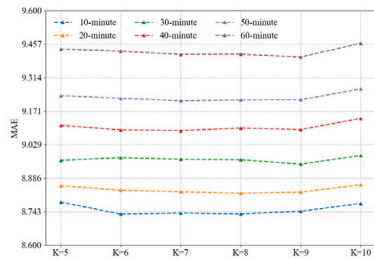
(a) MAEs of metro inflow prediction.



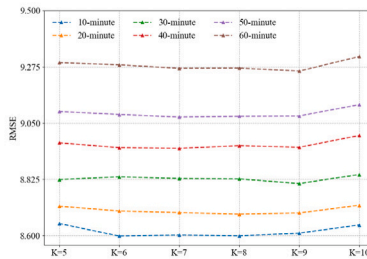
(b) RMSEs of metro inflow prediction.



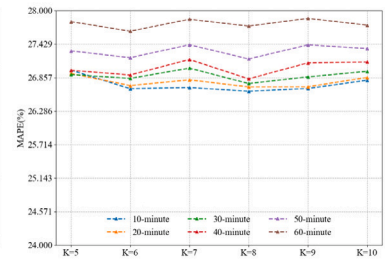
(c) MAPEs of metro inflow prediction.

Fig. 11. Performance of metro passenger inflow prediction when the cluster number of K equals different values.

(a) MAEs of metro outflow prediction.



(b) RMSEs of metro outflow prediction.



(c) MAPEs of metro outflow prediction.

Fig. 12. Prediction errors of metro passenger outflow of Fuzhou Metro when the cluster number of K equals different values.

(a) Passenger inflow KMSFC results.



(b) Passenger outflow KMSFC results.

Fig. 13. KMSFC results of Fuzhou Metro when $K = 8$.

Table 2

Performance comparison for all methods on Fuzhou Metro Passenger Inflow.

Metric	Time	HA	MLP	LSTM	GRU	GCN-SBULSTM	DCRNN	PVGCN	Graph WaveNet	Ada-STNet	DVRGCN
MAE	10 min	17.850	11.100	11.073	11.071	14.445	9.741	9.175	9.190	10.830	8.602
	20 min	19.894	11.612	10.708	10.799	14.670	10.096	9.410	9.418	12.007	8.832
	30 min	21.925	11.064	14.172	14.611	15.245	10.457	9.562	9.599	12.893	8.949
	40 min	23.971	11.272	13.390	14.370	15.648	10.937	9.756	9.832	13.477	9.093
	50 min	26.033	11.655	13.271	14.351	16.772	11.390	9.950	10.096	14.140	9.222
	60 min	28.071	12.306	13.294	14.389	18.186	11.961	10.108	10.474	14.988	9.331
RMSE	10 min	17.770	11.050	11.024	11.022	14.380	13.538	9.134	9.108	10.782	8.564
	20 min	19.805	11.560	10.660	10.750	14.604	14.231	9.368	9.334	11.953	8.792
	30 min	21.826	11.015	14.108	14.546	15.176	14.877	9.520	9.513	12.835	8.909
	40 min	23.863	11.221	13.330	14.305	15.578	15.749	9.712	9.744	13.416	9.052
	50 min	25.915	11.603	13.211	14.286	16.696	16.751	9.905	10.005	14.076	9.180
	60 min	27.944	12.251	13.235	14.324	18.103	18.035	10.063	10.380	14.920	9.289
MAPE (%)	10 min	42.003	35.979	30.230	30.111	40.068	27.008	25.654	25.549	29.454	23.914
	20 min	45.969	37.112	28.768	29.421	40.524	27.691	26.394	25.778	32.194	24.266
	30 min	50.067	29.826	31.041	31.920	35.716	28.172	26.627	26.948	34.397	24.489
	40 min	54.556	30.310	30.708	31.461	36.509	29.156	27.005	27.323	35.200	24.715
	50 min	59.608	30.708	30.593	31.308	38.235	29.989	27.545	27.263	36.097	25.019
	60 min	65.179	32.041	30.561	31.317	40.462	31.376	27.792	28.442	37.612	25.332

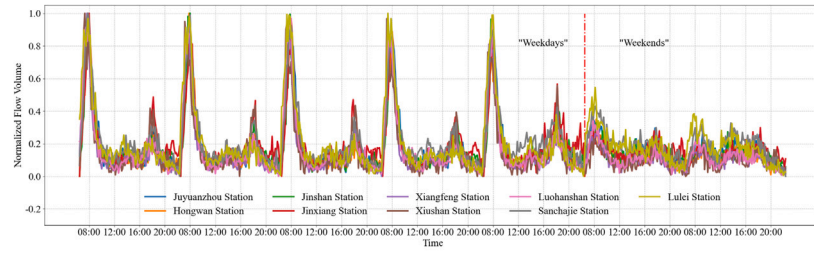
outflow prediction for the 20, 30, and 40-minute horizons, while MAEs and RMSEs of $K = 8$ are relative higher in inflow prediction. However, MAPEs of inflow and outflow predictions for the 20, 30, and 40-minute horizons are the lowest when $K = 8$. For the 60-minute horizon, the MAEs and RMSEs of both inflow and outflow predictions decrease from $K = 6$ to $K = 9$. Nevertheless, $K = 9$ performs poorly on MAPEs of the MPF predictions for the 60-minute horizon. When $K = 10$, the MAEs and RMSEs of both inflow and outflow predictions are high. Although MAPEs of $K = 10$ are less than $K = 9$ in metro inflow predictions, $K = 10$ obtains the overall worst performance. When the number of clusters ranges from 8 to 10 and the total amount of stations is only 42, there are some clusters which have only 1 or 2 stations. The main reason that the clustering algorithm generates such clusters is the particularity of certain station locations. For example, a cluster is formed to contain only Fuzhou Railway station and Fuzhou South Railway station because either metro station sits beside a railway station. Moreover, Gushan station is located at a tourism spot which displays a unique passenger flow variation. Based on the above experiments, the cluster number of K is set as 8 for the KMSFC in the comparison experiments.

The clustering results of metro passenger inflow and outflow are shown in Fig. 13(a) and 13(b), respectively. Those stations in the same cluster are denoted by triangles with the same color. Fig. 14 shows some clusters of metro stations, which are located around residential areas, workplaces, and railway stations. The clustering results shown in Fig. 14(a)–14(e) include the normalized MPF data in one week, which ranges from June 17th 2019 to June 23th 2019. The normalized MPF data for each station is adopted to express the flow changing tendency. In Fig. 14(a), metro stations correspond to light blue triangles in Fig. 13(a). The inflows of these metro stations increase dramatically around 8 a.m. since the residents nearby commute by metro for working. After two hours, the inflow volume goes down and stays at a low level. Although the inflows rise due to people getting off work at the evening rush hour, the increase is modest in comparison with the morning rush hour. Instead, the passenger outflows in Fig. 14(b) show the opposite trends. The outflows of stations keep low in the morning except Huangshan station and reach the peak at approximate 6 p.m. Commuters get off work and go back home around 6 p.m. which leads sudden increases of passenger outflows in these metro stations, as shown in Fig. 14(b). According to the commuting behaviors of urban residents and the fluctuations of inflows, the urban functional areas where metro stations sit (shown in Fig. 14(b)) are mainly residential areas. In addition, as shown in 14(a) and 14(b), there are some metro stations in both clusters but the inflow and outflow changing trends are totally different. Thus, the inflow and outflow patterns of either single metro station should be considered separately. In Fig. 14(c), the

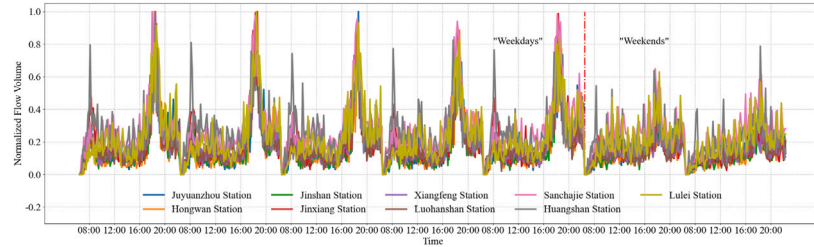
outflows hit the morning peak around 8 a.m. and the evening peak around 6 p.m. but the flow volume is much less in the evening peak. In the morning rush hours, a massive amount of commuters result in large outflows in these stations. The outflow pattern expresses that these stations are mostly surrounded by workplaces. Furthermore, there are many urban regions consisting of workplaces and residential areas. The metro stations shown in Fig. 14(d) have similar trends in the morning as the stations surrounded by residential areas, but the inflows jump significantly around 6 p.m. Beside, the peaks of inflows in the morning and evening are almost the same. The inflow pattern demonstrates that there are not only residential areas but also workplaces where metro stations are located (shown in Fig. 14(d)). By contrast, the weekday and weekend patterns of the metro stations linked to railway stations are different. As shown in Fig. 14(e), both inflows have obvious growth on Friday and Sunday afternoons. Thus, the KMSFC redivides “weekday” and “weekend” patterns in step one. The trends of inflows on Fridays and Sundays are relatively similar, and the weekend pattern of this cluster consists of Friday and Sunday. The weekday pattern involves the rest weekdays. The inflows of the metro stations connected to railway stations have similar fluctuations when those days are in the “weekday” pattern. In the “weekend” pattern, there is a great growth in the flow volume on Friday and Sunday evenings. Based on the passenger travel behavioral analysis, the inflow changes reflect the passenger travel patterns that the travelers from other cities get off the trains on Friday afternoons and stay for the weekends, while a great number of citizens return on Sunday afternoons after weekend breaks. From the result analysis of the KMSFC algorithm, the first step of the KMSFC can reconstruct the weekday and weekend patterns in a week for each metro station according to metro passenger travel patterns. Furthermore, the proposed KMSFC is capable to cluster the metro stations located at similar urban functional areas. The passenger travel patterns of metro stations in the same cluster are similar, and the graph constructed based on the KMSFC results contains more semantic information about the metro traffic network.

4.5.2. Experimental results: Comparison and analysis

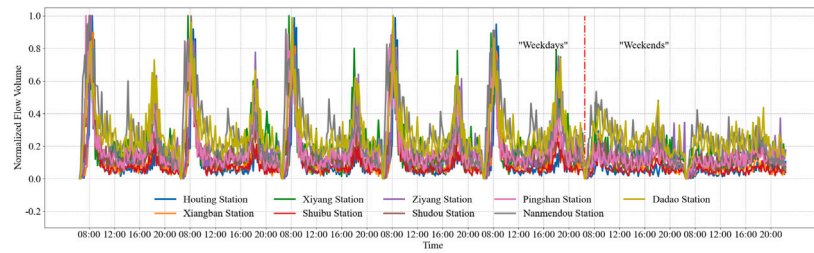
The method performance on Fuzhou metro passenger inflow (and outflow) prediction is shown in Table 2 (and 3). In general, the performances of most approaches declines when prediction horizon increases. From both tables, HA method produces unacceptable prediction errors. The main restriction of HA method is the assumption of stable traffic flow. However, traffic flow variations are usually unsteady in practical world. The DL-based methods can automatically learn nonlinear features of traffic data and the performances of three simple DL models (i.e. MLP, GRU and LSTM) are much better than the HA method.



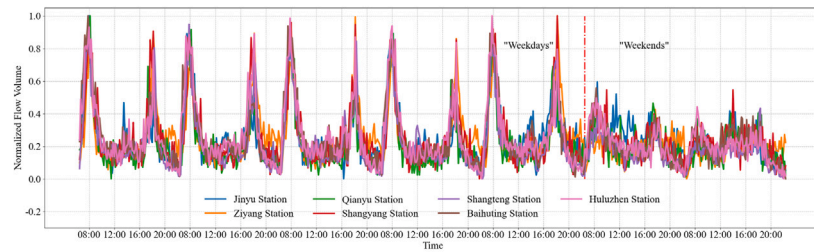
(a) Normalized passenger inflow of metro stations located at residential areas.



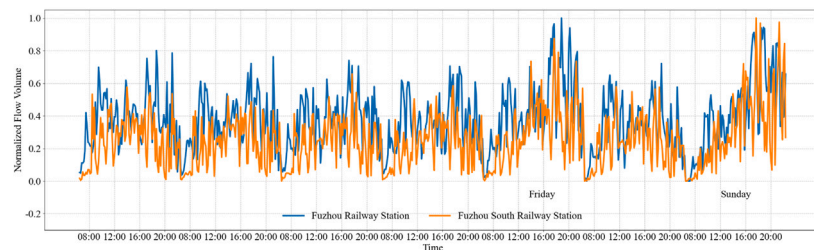
(b) Normalized passenger outflow of metro stations located at residential areas.



(c) Normalized passenger outflow of metro stations located at workplaces and residential areas.



(d) Normalized passenger inflow of metro stations located at workplaces and residential areas.



(e) Normalized passenger inflow of Fuzhou Railway station and Fuzhou South Railway station.

Fig. 14. Normalized MPF of Fuzhou metro stations located in similar urban functional areas.

Especially, the LSTM and GRU improve the MPF predictions for short-term (i.e. 10-minute and 20-minute) horizons due to their temporal

extraction ability. The GCN-SUBLSTM considers both spatial and temporal features of metro traffic dataset and improves the MPF prediction

Table 3

Performance comparison for all methods on Fuzhou Metro Passenger Outflow.

Metric	Time	HA	MLP	LSTM	GRU	GCN-SBULSTM	DCRNN	PVGCN	Graph WaveNet	Ada-STNet	DVRGCN
MAE	10 min	19.792	12.001	11.759	11.366	18.288	10.290	9.680	9.733	13.725	8.733
	20 min	21.370	12.261	11.154	11.057	17.372	10.489	9.705	10.003	12.978	8.821
	30 min	23.158	12.284	16.759	17.044	18.698	10.983	9.827	10.210	14.437	8.964
	40 min	24.887	12.500	16.127	16.832	17.731	11.275	9.987	10.494	14.939	9.099
	50 min	26.773	12.721	16.036	16.817	17.382	11.750	10.168	10.666	15.752	9.219
	60 min	29.579	13.008	16.025	16.818	17.566	12.428	10.365	11.095	17.956	9.414
RMSE	10 min	19.486	11.816	11.578	11.191	18.006	14.285	9.531	9.436	13.513	8.599
	20 min	21.040	12.072	10.983	10.887	17.104	14.655	9.557	9.698	12.778	8.686
	30 min	22.800	12.094	16.500	16.780	18.409	15.433	9.676	9.899	14.214	8.827
	40 min	24.502	12.307	15.878	16.572	17.458	16.097	9.834	10.173	14.708	8.960
	50 min	26.358	12.525	15.788	16.557	17.113	17.115	10.012	10.340	15.509	9.078
	60 min	29.121	12.807	15.778	16.558	17.295	18.294	10.207	10.756	17.678	9.270
MAPE (%)	10 min	57.743	45.026	35.415	35.124	56.395	32.277	30.994	32.365	37.967	26.623
	20 min	63.261	49.442	33.618	34.397	56.989	34.556	31.009	32.546	39.996	26.699
	30 min	69.454	37.036	39.347	41.402	49.913	37.231	31.772	33.673	43.727	26.755
	40 min	75.053	36.943	38.754	40.754	48.026	38.321	31.921	33.654	44.514	26.836
	50 min	80.936	38.241	38.560	40.495	47.987	39.012	32.148	34.343	47.350	27.176
	60 min	88.297	40.743	38.470	40.455	49.665	40.102	32.514	35.859	52.198	27.736

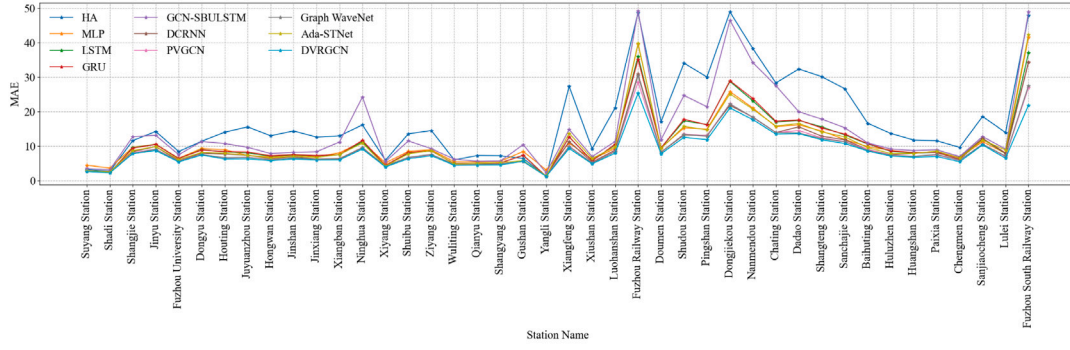
errors. The DCRNN and PVGCN both adopt the Seq2Seq architecture to capture the temporal features. Besides, the GCN-based spatial feature extraction modules are fused into the Seq2Seq architecture to recurrently capture the spatial features of the metro traffic network. Both the MAEs and MAPEs are greatly improved by the DCRNN, for example, the MAEs for the 10-minute horizons of inflow and outflow predictions are 9.741 and 10.290 respectively. However, the RMSEs of the DCRNN are higher than those simple DL models (i.e. MLP, LSTM, and GRU). Different from the DCRNN, the PVGCN takes advantage of comprehensive graphs which include more prior information of metro traffic network. The experimental results also demonstrate that the PVGCN outperforms above mentioned baselines on the MPF predictions for all time horizons. The graph waveNet and Ada-STNet build independent temporal feature extraction modules along with the GCN modules to sequentially capture spatiotemporal features. Unlike other GCN-based baseline methods, their GCN modules are based on the self-adaptive adjacency matrix and static adjacency matrix. From the experimental results, the graph waveNet has more obvious improvements on the MPF prediction than Ada-STNet. From 10-minute to 30-minute horizons, the graph waveNet achieves outstanding performance on the RMSEs of inflow prediction which are 9.108, 9.334, and 9.513. Moreover, the RMSEs of the outflow prediction at 10-minute and 20-minute produced by the graph waveNet are lowest among baselines. Among all baselines, both the PVGCN and graph waveNet effectively capture the spatiotemporal information of the metro traffic network. Moreover, the PVGCN performs better than the graph waveNet in the metro passenger outflow among all baselines except the RMSEs for the short-term horizons.

The last columns of Tables 2 and 3 display the MAEs, RMSEs, and MAPEs of the MPF predictions generated by the DVRGCN. It is apparent that the proposed DVRGCN achieves the overall best performance among all the compared methods. Compared to the simple methods (i.e. HA, MLP, LSTM and GRU), the DVRGCN has an more intricate structure, which includes the GCN modules and Seq2Seq architecture, to capture the spatiotemporal features of metro data. Among the simple methods, the DVRGCN decreases by at least 19.12%, 17.51%, and 15.65% of the MAE, RMSE, and MAPE in the inflow prediction and the DVRGCN improves at least 20.22%, 20.21%, and 20.58% of the MAE, RMSE, and MAPE in the outflow prediction, respectively, for six different time horizons. Other baselines adopt diverse GCN-based spatial and temporal feature extraction approaches to capture spatiotemporal patterns of traffic data. As shown in Table 1, the GCN-SBULSTM, DCRNN and PVGCN only construct static matrices with spatial information of traffic network for effectively extracting local spatial features. The graph waveNet and Ada-STNet adopt both self-adaptive adjacency matrix and simple adjacency matrix to capture

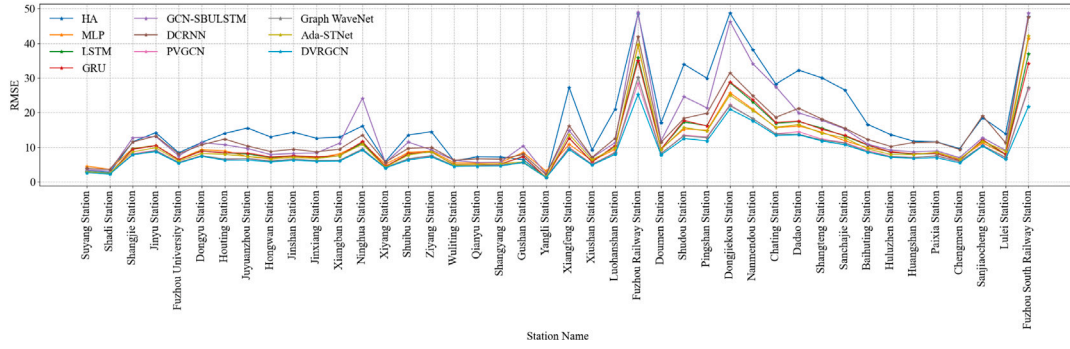
more comprehensive spatial features. However, they neglect the information of similar passenger travel patterns among all metro stations. The proposed KMSFC makes up for this deficiency by providing groups of stations within the same urban functional area. The DVRGCN merges the advantages of both kinds of GCN-based method to exploit static and dynamic spatial patterns of the metro traffic network. In addition, all of baselines fail to take influences of external features into account. For all time horizons of metro inflow predictions, the DVRGCN exhibits the MAEs, RMSEs, and MAPEs reductions of 6.142%, 5.802% and 5.867%. Moreover, the outflow predictions are also improved by the DVRGCN at least 8.778%, 8.776% and 13.90% on the MAEs, RMSEs, and MAPEs respectively.

In the benchmark tests, most methods perform not well on Fuzhou Railway station, Fuzhou South Railway station, and Dongjiiekou station because the MPFs of these stations are large and passenger behavior patterns are extremely intricate. Based on the KMSFC results, two railway stations are in the same cluster and Dongjiiekou station is in another cluster which also contains other shopping centers. The pre-defined graph successfully encodes these prior spatial information for the DVRGCN to make accurate predictions. For the MAPEs, all methods produce much higher on Suyang, Shadi, Gushan, Yangli and Fuzhou South Railway stations than others. Except Fuzhou South Railway station, the MPFs of other stations are very small since these stations are located in relatively remote areas. In other words, the MPF predictions on these stations are unstable. Compared to other baselines, the proposed DVRGCN produces the lowest errors on these stations and has a stable performance on the MPF predictions of all stations.

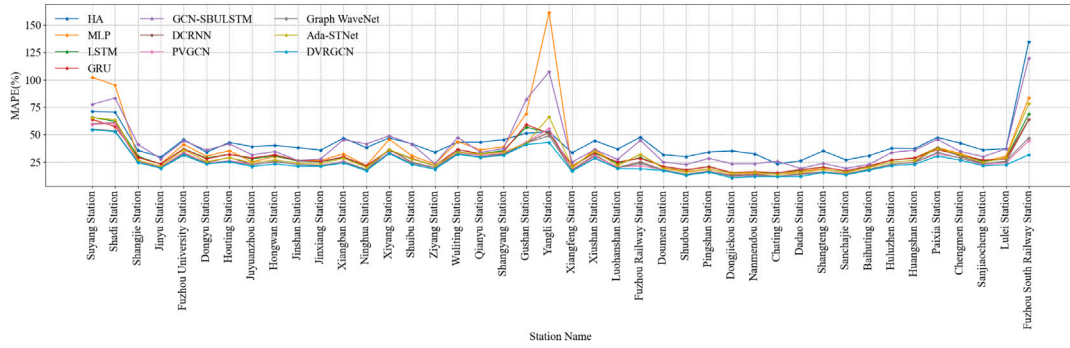
The prediction errors of all metro stations for the 10-minute horizon are shown in Fig. 15(a)–15(c). For the MAEs and RMSEs, those stations with higher MPF volumes cause larger prediction errors. Meanwhile, the MAPEs of some stations jump extremely high because the original MPFs are very small. Based on the experimental results, the proposed KMSFC provides valuable spatial relations among metro stations and DVRGCN is effective in capturing the spatiotemporal features of metro traffic data and making stable MPF predictions. Figs. 16 and 17 show the metro passenger inflow and outflow predictions of Dongjiiekou station from June 24th 2019 to June 30th 2019. Dongjiiekou station locates in the biggest business center of Fuzhou and is the busiest station in Fuzhou metro system. The MPF has violent fluctuations in both morning and evening rush hours on weekdays and DVRGCN responses more quickly to the sudden increase and decrease of MPF than other methods. On weekends, amounts of people spend their afternoon and evening time on the bustling business center. Therefore, the inflow of Dongjiiekou station increase gradually and the outflow remains at a high level. As shown in Figs. 16 and 17, passenger travel patterns on both weekdays and weekends are precisely captured by DVRGCN.



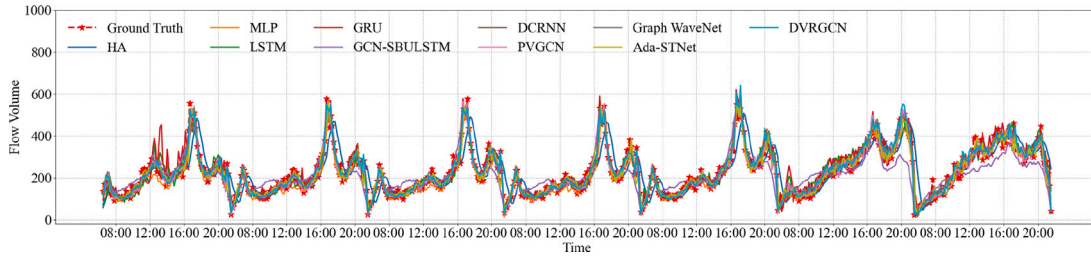
(a) MAEs of passenger inflow predictions on 10-minute.



(b) RMSEs of passenger inflow predictions on 10-minute.



(c) MAPEs of passenger inflow predictions on 10-minute.

Fig. 15. Experimental results of metro passenger inflow predictions on 10-minute.**Fig. 16.** Passenger inflow predictions on Dongjiekou station.

4.6. Ablation experiments

To identify contributions of different modules in DVR-GCN, the ablation experiments are conducted on the same metro traffic dataset. There are six variants of the DVRGCN for comparison as follows:

(1) *DVRGCN_{w/o-AGCN}*: DVRGCN removes A-GCN from GRSU and only uses GRU-MGCN in GRSU.

(2) *DVRGCN_{w/o-GRU-MGCN}*: DVRGCN removes GRU-MGCN from GRSU and only uses AGCN.

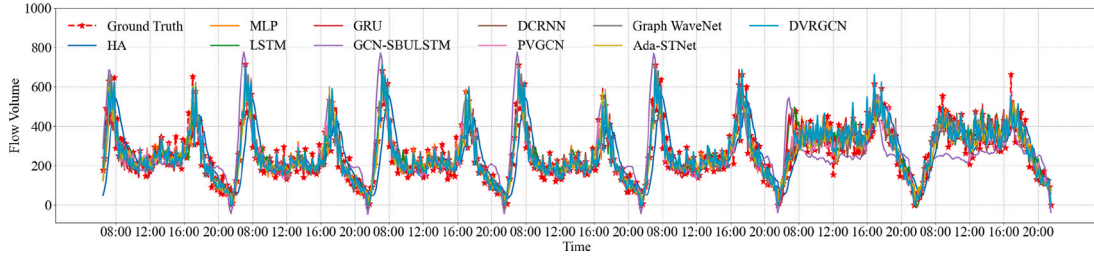
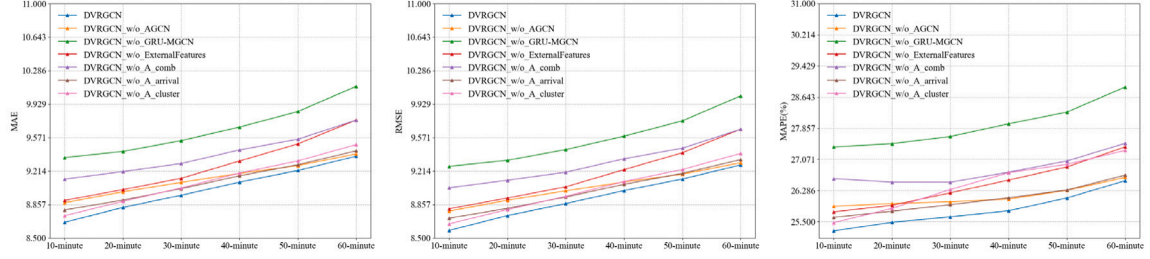
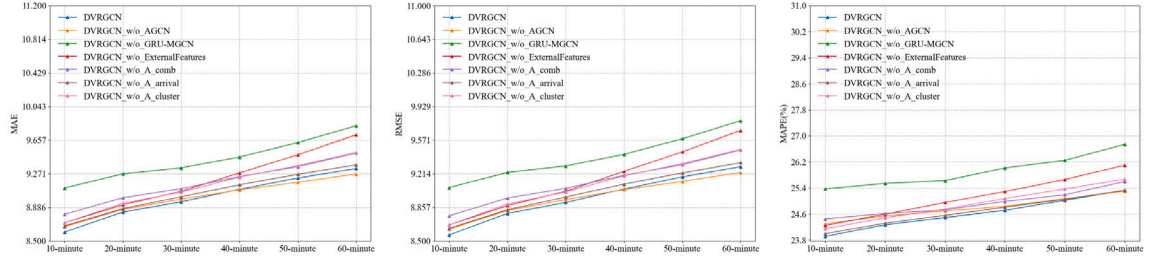


Fig. 17. Passenger outflow predictions on Dongjiekou station.



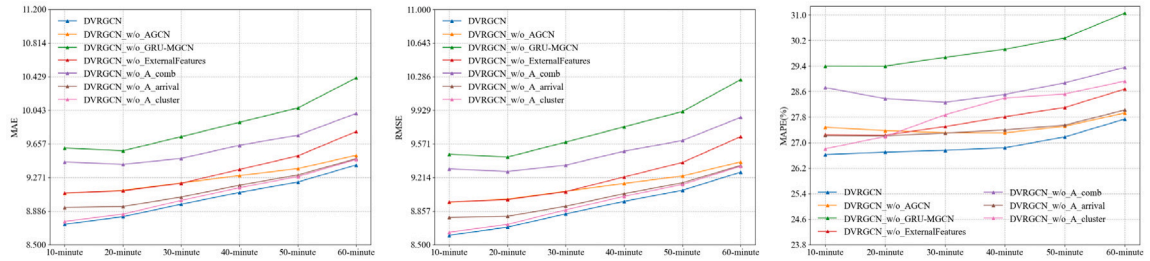
(a) MAEs of MPF predictions - ablation experiments. (b) RMSEs of MPF predictions - ablation experiments. (c) MAPEs of MPF predictions - ablation experiments.

Fig. 18. Results of ablation experiments.



(a) MAEs of metro inflow predictions - ablation experiments. (b) RMSEs of metro inflow predictions - ablation experiments. (c) MAPEs of metro inflow predictions - ablation experiments.

Fig. 19. Results of ablation experiments.



(a) MAEs of metro outflow predictions - ablation experiments. (b) RMSEs of metro outflow predictions - ablation experiments. (c) MAPEs of metro outflow predictions - ablation experiments.

Fig. 20. Results of ablation experiments.

(3) *DVRGCN_w/o_ExternalFeatures*: External Features denoting the implicit MPF features are removed from DVRGCN.

(4) *DVRGCN_w/o_A_comb*: To verify the effectiveness of A^{comb} in (11), DVRGCN takes geographically adjacency matrix for MPF prediction.

(5) *DVRGCN_w/o_A_arrival* (without arrival adjacency matrix $A^{arrival}$): To verify the effectiveness of $A^{arrival}$ in (11), DVRGCN takes only $A^{cluster}$ for MPF prediction.

(6) *DVRGCN_w/o_A_cluster* (without KMSFC-based adjacency matrix $A^{cluster}$): To verify the effectiveness of $A^{cluster}$ in (11), DVRGCN takes only $A^{arrival}$ for MPF prediction. Fig. 20 shows the MAEs, RMSEs, and MAPEs of the MPF predictions of the DVRGCN and its variants. Fig. 18(a)–18(c) present the average errors of inflow and outflow predictions, while Fig. 19(a)–19(c) and Fig. 20(a)–20(c) display all three metrics of inflow and outflow predictions, respectively. In all plots, blue lines indicate the errors obtained by the original DVRGCN. Orange lines in all figures represent the errors generated by *DVRGCN_w/o_AGCN*.

Table 4

Training time and parameter quantity of all implemented methods.

Methods	HA	MLP	LSTM	GRU	GCN-SBULSTM	DCRNN	PVGCN	Graph WaveNet	Ada-STNet	DVRGCN
Time	0 s	0.38 s	0.42 s	0.41 s	0.82 s	25.48 s	11.93 s	2.38 s	12.36	18.00 s
#_Param	0M	0.324M	0.898M	0.679M	9.716M	0.373M	22.61M	0.303M	0.832M	18.64M

As shown in Fig. 18(a)–18(c), all orange lines are above blue lines original DVRGCN. Concretely, the MAE and RMSE of the inflow predictions generated by *DVRGCN_w/o_AGCN* are lower than the original DVRGCN after 40 min and the MAPE is lower for the 60-minute horizon. However, the original DVRGCN achieves much better outflow prediction results compared to *DVRGCN_w/o_AGCN*. Generally, the DVRGCN benefits from hidden spatial features learning ability of the AGCN module. *DVRGCN_w/o_GRU – MGCN* denoted by green lines obtain the overall worst performance. The GRU-MGCN is the core module to extract the local spatial features of the metro traffic network. The lack of the GRU-MGCN leads to unacceptable errors of MPF predictions and the significance of the GRU-MGCN is obvious. *DVRGCN_w/o_ExternalFeatures* in red show that the absence of external features, which include time and flow volume features, causes the increases of errors. Especially, the MPF predictions after 30-minute horizon become worse when there are no fusion of external features. Sudden increases of errors indicate that the fusion of external features make the DVRGCN better model the MPF changing patterns at critical time points of different metro stations. Purple lines in all figures represent the errors produced by *DVRGCN_w/o_A_comb*. As shown in Fig. 18(a)–18(c), the performance of *DVRGCN_w/o_A_comb* degrades a lot due to the loss of semantic information of metro traffic network. Especially for the errors of outflow predictions shown in Fig. 20(a)–20(c), three metrics produced by *DVRGCN_w/o_A_comb* are much higher than original model.

To further compare the performance of $A^{cluster}$ and $A^{arrival}$, experiments are conducted on variants *DVRGCN_w/o_A_arrival* and *DVRGCN_w/o_A_cluster*. In the comparisons of three metrics in 18(a)–18(c), gaps between brown and blue lines, which indicates *DVRGCN_w/o_A_arrival* and *DVRGCN*, are gradually narrowing. Especially, outflow prediction errors shown in Fig. 20(a)–20(c) reconfirm this view. The narrowing of the gap illustrates that $A^{arrival}$ with valuable information makes *DVRGCN* performing better in short-term MPF forecasting. When removing $A^{cluster}$ from A^{cmob} , *DVRGCN_w/o_A_cluster* does not achieve excellent performance, particularly MAPEs of outflow predictions (shown in Fig. 20(c)) after 30-minute horizon. From 18(a)–18(c), *DVRGCN_w/o_A_cluster* denoted by pink lines expresses overall inferior performance on MPF predictions. Therefore, *DVRGCN* benefits from $A^{cluster}$ since the KMSFC provides rich spatial relation information of the metro traffic network. In summary, as shown in Fig. 18(a)–18(c), the original DVRGCN in blue lines outperforms its variants on the MPF predictions for all the six horizons. According to ablation experimental results, different modules have made contributions and brought valuable information for the DVRGCN to make precise MPF predictions.

5. Conclusions and future work

In this paper, a novel metro passenger flow prediction method is proposed. There are three parts in the proposed method which are the KMSFC, external feature fusion and DVRGCN. In the KMSFC algorithm, the two-step clustering discovers the weekday and weekend variation patterns of the MPF and those metro stations located at similar urban functional areas are clustered. The combined matrix A^{comb} based on the clustering results and arrival time contain plenty of spatiotemporal information. In addition, this study also concerns the external features which reflect the MPF variation features during significant time periods and in different metro stations. Finally, the

DVRGCN recurrently extracts the spatiotemporal features of the metro traffic networks and precisely predicts the future MPF. Compared with the state-of-the-art traffic prediction techniques, the proposed metro passenger prediction methods are superior on the practical metro traffic dataset. The ablation experiments conducted on the metro data prove the contributions of different components in the DVRGCN. In the future, the KMSFC algorithm can be adapted for other traffic pattern clustering demands such as highway traffic and urban road traffic. Currently, the powerful GPUs enable large DL training but saving computing resources is also important. As the computation time and parameter quantity explicitly demonstrate the computation resource consumption, one epoch average training time and parameter quantity of all the compared methods are given in Table 4. Although the proposed method achieves the best performance, the computation time and parameter quantity required by the DVRGCN are both the second most among those methods. Therefore, how to simplify the architecture and calculations of the proposed method without sacrificing the performance would be a critical research direction in the future.

CRedit authorship contribution statement

Hao Fang: Conceptualization, Methodology, Software, Writing - Original draft. **Chi-Hua Chen:** Formal analysis, Validation, Data curation, Resources, Writing - Original draft, Funding acquisition. **Feng-Jang Hwang:** Formal analysis, Validation, Investigation, Writing - Review & Editing. **Ching-Chun Chang:** Visualization, Writing - Review & Editing. **Chin-Chen Chang:** Supervision, Writing - Review & Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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